

The Dollar's Safe-Haven Status Across Two Shocks: Evidence from Liberation Day and the Strait of Hormuz Crisis

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1 Introduction

Within a single year, two major shocks moved global commodity and financial markets. On 2 April 2025, widely referred to as “Liberation Day”, the United States government announced broad-based tariffs on virtually all of its trading partners, citing persistent trade deficits and national security concerns. The policy included a base 10 percent tariff and, subsequently, higher rates for trading partners with large surpluses against the United States, including China and the European Union. Following the announcement, the dollar fell sharply against a broad basket of currencies, with the euro, yen and Swiss franc appreciating against it. The move was accompanied by a spike in equity-market volatility and a U.S. equities sell-off.

Less than a year later, on 28 February 2026, the United States and Israel launched airstrikes on Iran, beginning the 2026 Iran war. As escalation in the Gulf threatened to close the Strait of Hormuz and sent crude prices sharply higher, the dollar appreciated. Both episodes involved the United States, yet the dollar moved in opposite directions.

This thesis speaks to a live debate about whether the dollar’s exceptional status is eroding. The dollar’s fall during the tariff episode was unusual because it coincided with rising market volatility and rising Treasury yields, conditions under which the dollar has often strengthened ([Kamin 2025](#); [Jiang et al. 2026](#)). Earlier research documents the dollar’s tendency to appreciate during episodes of heightened global risk aversion ([Habib and Stracca 2012](#); [De Bock and de Carvalho Filho 2015](#)). The tariff response raises the possibility that this protection weakens when U.S. policy itself unsettles markets. One episode cannot establish a lasting loss of safe-haven status, however. The subsequent Hormuz crisis offers a contrasting case in which the United States was involved in a military and oil-supply shock.

The central question is whether the dollar’s safe-haven response depends on the type of shock, even when the United States is involved in both. The analysis compares the U.S.-originated Liberation Day tariffs with the 2026 Israel–Iran/Hormuz crisis, in which the United States took part as a co-belligerent. Russia’s invasion of Ukraine in February 2022 provides an external military and oil-supply benchmark without direct U.S. participation. If the two military shocks show similar dollar responses while the tariff shock differs, the pattern is consistent with shock type playing an important role, though the comparison cannot establish a general causal effect of shock type.

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The empirical analysis uses daily data from January 2019 to June 2026. It combines a panel of 32 spot exchange rates from the London Stock Exchange Group (LSEG) with one-month forward quotes for 28 currencies and data on gold, equities and crude oil. The main method is an event study that measures cumulative abnormal returns around each shock for the broad dollar index, currency baskets, gold, equities and crude oil. Formal cross-event tests compare the dollar's abnormal returns. As a robustness check, a regression examines whether the dollar's usual relationship with market risk changes across the episodes.

The contribution is a comparison of the Liberation Day and Hormuz episodes within one consistent empirical design. The currency baskets distinguish funding and carry currencies, established haven alternatives, and oil exporters and importers. This allows the dollar's response to be read alongside the currencies most exposed to changes in risk appetite and crude prices. Gold offers another benchmark. Examining gold and oil in both dollars and euros also shows whether a finding depends on the currency in which the commodity price is expressed. These comparisons connect the safe-haven and commodity-currency literatures without treating the dollar's movement in isolation.

The results suggest that the dollar's safe-haven behaviour varies across the shocks. In the first two trading days of each episode, the broad dollar index had a cumulative abnormal return of -1.44 percent for Liberation Day and $+1.39$ percent for the Hormuz crisis. At the twenty-day horizon, the contrast grows, although price movements over that period may also reflect other developments. The Ukraine benchmark also shows dollar appreciation at the five-day horizon. Regressions using market-based risk measures support the event-study pattern in their point estimates. The dollar's normal positive sensitivity to risk disappears during the tariff window but remains positive during the military episodes. Gold provides mixed evidence for a universal safe-haven role because its response depends on the episode, horizon and denomination. The clearest separation between oil exporters and importers occurs after Liberation Day, when crude prices fell, while the Hormuz spread remains insignificant.

The remainder of the thesis is organised as follows. Chapter 2 reviews the safe-haven, dollar-erosion and commodity-currency literatures. Chapter 3 sets out the conceptual framework, including the working definition of a safe-haven currency, a classification that separates a shock's origin from its nature, and the hypotheses that follow. Chapter 4 describes the events. Chapter 5 describes the data and the event-study and regression methodologies. Chapter 6 presents the empirical results.

1 Introduction

Chapter 7 discusses the findings and their implications, and Chapter 8 concludes.

2 Literature Review

This chapter situates the thesis within six distinct strands of literature. Each body of research addresses a specific dimension of the core question the thesis tries to answer: does the dollar’s status as a safe haven depend on the nature of a shock, rather than its source? By reviewing these fields, this chapter establishes the theoretical framework to understand how the dollar functions as a safe haven, setting the stage for examining the consequences of when the United States itself originates a shock, acts as a co-belligerent or is not directly involved.

2.1 Currency Risk Factors

[Lustig et al. \(2011\)](#) establish the foundational framework for understanding currency risk by identifying two primary common risk factors, a “level” factor, the average excess return on foreign-currency portfolios against the dollar, and a “slope” factor, which captures the excess returns of carry trades. They demonstrate that high-interest-rate currencies load more heavily on the slope factor, which is closely linked to global equity-market volatility. Crucially, their model shows that during periods of high global risk, low-interest-rate currencies endogenously appreciate, acting as a hedge, while high-interest-rate currencies depreciate.

The dollar’s special role is not confined to asset markets. It also dominates the invoicing of international trade. Drawing on evidence from more than 2,500 country pairs covering some ninety percent of world trade, [Gopinath et al. \(2020\)](#) show that traded-goods prices are predominantly sticky in dollars rather than in the exporter’s or the importer’s currency. Trade prices and volumes therefore respond to the dollar exchange rate more than to the bilateral rate.

Building on this dominant-currency paradigm, [Dalgic and Ozhan \(2025\)](#) show that a currency’s risk premium is not exogenous but rooted in an economy’s trade and financial structure, namely the share of its trade invoiced in dollars and the amount of dollar debt it carries. In their small open-economy model, a shock that depreciates the local currency, such as a foreign interest-rate hike, turns contractionary rather than expansionary. Dollar debt opens a financial channel through which the depreciation raises the real value of liabilities. Servicing dollar debt now requires more local currency. Dollar invoicing mutes the trade channel, leaving export volumes largely unresponsive to the exchange rate, providing no offset. Because depreciations are therefore recessionary, the exchange rate turns countercyclical. The currency loses

value precisely when domestic income falls, offers its holder no insurance, and is therefore a poor hedge against domestic downturns. Investors hold such currencies only against a sufficient risk premium, the mirror image of the safety commanded by the dollar. The dollar and carry factors, extended to June 2026, form the basis of the portfolio construction in Chapter 5.

2.2 Dollar Dominance and Its Erosion

The dollar’s role as the primary global reserve currency confers an “exorbitant privilege”, a persistent excess return on U.S. external assets over its liabilities (Gourinchas and Rey 2005). [Gourinchas et al. \(2017\)](#) show that this privilege is mirrored by an “exorbitant duty”. In global crises the United States transfers wealth abroad, roughly 19% of U.S. gross domestic product (GDP) during 2007–2009. Because it holds risky foreign assets while owing safe dollar liabilities, collapsing global asset prices and the appreciating dollar both push its net foreign asset position downward, with the exchange rate accounting for roughly a third of the valuation effect. [Jiang et al. \(2021\)](#) find the source of that movement, showing that surges in foreign demand for safe dollar assets appreciate the dollar immediately. They capture the demand through the convenience yield on U.S. Treasuries, the return investors give up in exchange for safety and liquidity. The dollar’s appreciation in a global crisis is therefore a consequence of its role as the world’s supplier of safe assets. [Obstfeld and Zhou \(2023\)](#) add the cyclical dimension, describing a “global dollar cycle” in which dollar appreciation tightens financial conditions worldwide. They find that dollar movements correlate most reliably with measures of investor risk appetite.

While the 2008 financial crisis and the COVID-19 pandemic elicited the standard appreciation, [Jiang et al. \(2026\)](#) argue that the dollar has been losing that standing gradually, documenting convenience yields on dollar safe assets eroding over the two years before the 2025 tariff announcements. A comparable trend is evident in central bank reserves. [Arslanalp et al. \(2022\)](#) document a two-decade decline in the dollar’s share of global reserves, and attribute it to deliberate diversification by reserve managers rather than to valuation effects. Between 1999 and 2021 the dollar’s share of global reserves declined from 71% to 59%. A quarter of what left the dollar moved into the Chinese renminbi, the remaining three quarters into currencies of smaller economies that had rarely served as reserve assets before.

2.3 Safe-Haven Currencies and Gold

A central paradox, identified by [Habib and Stracca \(2012\)](#), is the dollar's appreciation during the 2008 global financial crisis, a crisis that originated in the United States. Analysing 52 currencies over almost 25 years, they find that safe-haven status is best predicted not by the interest-rate spread that the carry-trade literature emphasises but by a country's net foreign asset position, which signals how exposed it is to external stress. A currency that has hedged well in the past also tends to continue doing so, and the currencies of large but less financially open economies hedge better in periods of distress. The interest-rate spread against the United States, by contrast, matters only during the 2008 crisis itself.

[Cheema et al. \(2022\)](#) compare a range of safe-haven assets across different crises, and find their safe-haven character changed considerably during the COVID-19 pandemic compared with the 2008 financial crisis. The Swiss franc remained an intermediate haven in most settings, while gold and silver, weak havens in 2008, turned risky. The dollar deteriorated as well, from intermediate protection in 2008 to weak protection during the pandemic.

Providing broader context, [De Bock and de Carvalho Filho \(2015\)](#) demonstrate that the dollar, Japanese yen and Swiss franc consistently appreciate against other currencies during risk-off episodes. How much each currency gains depends on its interest rate before the episode, its net foreign asset position, whether it is overvalued, and how cheaply and freely it can be traded. [Ranaldo and Söderlind \(2010\)](#) show that during U.S. equity sell-offs the franc and yen appreciate against the dollar, with effects that appear within hours, persist for several days, and grow more than proportionally as volatility rises. The dollar therefore strengthens broadly in risk-off episodes while still losing ground to the franc and the yen.

[Baur and Lucey \(2010\)](#) are the first to formally define and test the distinction between a hedge and a safe haven. In their terminology, a hedge is uncorrelated or negatively correlated with equities on average, which does not guarantee protection during market crashes. A safe haven must deliver exactly that protection in the extreme. Applying the distinction to gold, they find that it hedges equities on average, but works as a safe haven only after severe market downturns, and even then only for a short time period. This strand supplies the thesis's working definition of a safe haven and its benchmarks. Chapter 3 adopts the conditional, crisis-contingent notion of safety, and the yen, the franc and gold serve as the standards against which the dollar's response is measured.

2.4 Geopolitical Risk

Caldara and Iacoviello (2022) provide the foundational measurement of geopolitical risk, a news-based index that separates geopolitical “threats”, events that are feared but have not yet occurred, from geopolitical “acts”, their actual realisation. Shocks to the index depress investment, employment and stock prices, and the damage persists, with both the threat and the realisation components contributing. Expanding this to exchange rates, Yilmazkuday (2025) demonstrates that the dollar’s geopolitical safe-haven status is not universal. A year after a shock to global geopolitical risk, the U.S. dollar has depreciated significantly, even as currencies such as the South African rand or Brazilian real have appreciated. The currencies of countries most deeply integrated into international supply chains depreciate most, a pattern driven primarily by geopolitical acts rather than threats. Complementing this, Liu and Zhang (2024) find that a zero-cost strategy that buys the currencies of high-geopolitical-risk countries and sells those of low-risk countries earns a significant excess return of 5.72% per year, and that this return predictability stems from country-specific and regional risk components rather than a single global factor. A currency’s risk–return profile is therefore sensitive to the geographic origin of the shock.

Institutional evidence points in the same direction. The International Monetary Fund (2025b) finds that although geopolitical risk events exert only a modest effect on asset prices on average, major events, military conflicts in particular, produce substantially larger and more persistent equity declines and raise governments’ borrowing costs. The effects are strongest in emerging markets with little fiscal room and limited foreign reserves, and they spill over to other countries through trade and financial ties.

The thesis draws on this strand directly. The aggregate index places the Hormuz and Ukraine episodes in their longer-run context in Chapter 4, and changes in the index serve as a measure of geopolitical risk in the safe-haven regression.

2.5 Tariff Shocks and Exchange Rates

The 2025 tariff episode has generated a fast-growing literature that is central to this thesis. Kamin (2025) estimates the dollar’s daily relationship with the VIX, the Chicago Board Options Exchange’s index of expected S&P 500 volatility, and with interest differentials and finds a significant break after the Liberation Day announcements of April 2025. Its sensitivity to the VIX turns from positive to

negative, so a currency that once gained when volatility spiked now falls with it, and investors no longer treated the dollar as a refuge in stress. [Jiang et al. \(2026\)](#) document the same break across a wider set of correlations. The dollar depreciated while U.S. interest rates rose relative to euro rates, the VIX climbed and the convenience yield on Treasuries fell, a pattern they attribute to weakening foreign demand for the safe assets the United States supplies. [Hassan et al. \(2025\)](#) provide the theoretical counterpart in a general-equilibrium model where a currency's safety arises endogenously. In their account the dollar is safe because disturbances originating in the United States move traded-goods prices worldwide, and a trade war that cuts the country off from world markets shuts down those spillovers, so the dollar's habit of strengthening in bad times fades and U.S. interest rates move higher.

The depreciation was not what the earlier record suggested. [Khalil and Strobel \(2024\)](#) show that the tariff rounds of 2018–2019 came with a stronger dollar and trace that appreciation to U.S. trade policy itself. [Ostry et al. \(2025\)](#) qualify the pattern with a database of tariff announcements, threats and implementations spanning 2018–2020 and 2025, weighting each shock by the size of the tariff and its economic relevance. The direction of the response, they find, hinges on retaliation. The dollar appreciates when the United States imposes tariffs unilaterally but depreciates when trading partners strike back, which leads them to conclude that the fall after 2 April 2025 was no anomaly once retaliation is taken into account. [Kalemli-Özcan et al. \(2026\)](#) offer a complementary explanation centred on uncertainty. In their model, doubt about future tariff policy feeds a risk premium into the exchange rate, and greater tariff uncertainty raises precautionary saving and the compensation investors demand, weakening the currency at once even though the tariffs alone would strengthen it. Calibrated to the 2025 episode, the model matches how far and how fast the dollar fell.

[Benguria and Saffie \(2025\)](#) measure the immediate market response to the 2 April announcement itself, separating the anticipated part of each country's tariff from the surprise created by rounding the rates upward. A one-percentage-point higher surprise tariff lowers the local-currency equity prices of listed firms in the affected trading-partner countries by 0.23%, and the anticipated part also lowers them, significantly but by an order of magnitude less. The dollar shows no appreciation, if anything a depreciation against floating currencies. This strand supplies the thesis's first treatment. The Liberation Day shock enters the event study as the trade-policy episode against which the military and oil-supply shocks are compared.

2.6 Commodity Currencies and Oil

Y.-C. Chen and Rogoff (2003) show that for commodity exporters such as Australia, Canada and New Zealand, the world price of their exports is a strong driver of the real exchange rate. These three serve as the natural test cases, developed economies with floating exchange rates whose exports remain dominated by commodities. The effect is clearest for the Australian and New Zealand dollars. Kilian (2009) decomposes oil-price changes into supply disruptions, global demand for industrial commodities, and precautionary demand reflecting concern about future supply, each with distinct effects on the oil price, output and inflation. A supply-driven price increase is therefore not equivalent to a demand-driven one.

H. Chen et al. (2016) show that the dollar reacts more strongly to demand-driven than to supply-driven oil shocks. The reaction is largest against the Canadian dollar and the Norwegian krone, because Canada and Norway trade oil heavily with the United States. Oil has also become more important for the dollar over time. Before the global financial crisis, oil shocks accounted for 10% to 20% of the movement in the dollar's bilateral rates. Afterwards the share rises to over 40%, and for the Canadian dollar to more than half.

Albulescu and Ajmi (2021) document predictive causality between oil prices and the dollar in both directions, strongest in crisis periods. Salisu et al. (2022) show that oil tail risk predicts the tail risk of the dollar's rates against the Canadian dollar and the British pound, while the dollar-yen pair moves inversely, consistent with the yen's haven role.

For the thesis, this literature provides the basis for the oil-exporter versus oil-importer portfolio split in Chapter 5, and the distinction between supply-driven and demand-driven oil movements (Kilian 2009) separates the Hormuz crisis from the tariff shock. The episodes themselves, and the sources that document them, are covered in Chapter 4.

3 Conceptual Framework

3.1 The Dollar as a Safe Haven

In periods of high global risk investors reallocate from risky assets toward safe assets, meaning assets they expect to hold value. Global risk is high when equity markets fall and expected volatility rises. For the dollar this demand concentrates in U.S. Treasuries, whose safety and liquidity command a premium. The resulting surge in demand for dollar safe assets appreciates the dollar ([Jiang et al. 2021](#)).

The safe-haven channel connects two observable variables. Global risk raises the demand for safe assets and is measured with the VIX. The dollar's return shows whether that demand appreciates the dollar and is measured on the broad dollar index and the currency baskets of Chapter 5. Following [Baur and Lucey \(2010\)](#), an asset is a hedge if it is uncorrelated or negatively correlated with risky assets on average, and a safe haven if that protection holds in times of market stress. The dollar therefore behaves as a safe haven if its return rises when equity prices fall and volatility increases ([Ranaldo and Söderlind 2010](#); [De Bock and de Carvalho Filho 2015](#)).

The channel rests on three assumptions. First, exchange rates incorporate public news within days, so a short event window can attribute a price move to the event. Second, the VIX is a valid measure of global risk appetite. Third, around large shocks the dollar's short-window response reflects the demand for safety rather than changing interest-rate expectations. Under these assumptions the channel makes a sign prediction. In the days after the announcement of a large adverse shock, the dollar's cumulative abnormal return from day 0 to day h , $CAR(0, h)$ as defined in Section 5.2, should be positive,

$$CAR(0, h) > 0 \quad \text{for short horizons } h.$$

The event study of Chapter 5 tests exactly this sign. The same logic implies that on ordinary days the dollar's return should rise with the VIX, a positive relation the regression checks as supporting evidence. The origin of a shock appears nowhere in the channel. On its own it therefore predicts a positive abnormal return around every risk-off event, even one that originates from U.S. policy itself.

3.2 The Limits of Dollar Safety

The dollar's value includes a premium for safety. Investors accept a lower return on U.S. safe assets, the convenience yield, and this premium is priced into the dollar's exchange rate (Jiang et al. 2021). The premium rests on the expectation that dollar assets will remain safe. A shock that originates in U.S. policy can lower exactly this expectation. The demand for dollar safe assets then falls despite rising global risk, and the dollar depreciates (Hassan et al. 2025). Around a U.S.-originated trade-policy shock this erosion channel therefore predicts the opposite sign on the same quantity, a negative cumulative abnormal return in the days after the announcement.

The erosion channel runs through the same two variables as the safe-haven channel, global risk and the dollar's return, and adds the safety premium as a third. It rests on two assumptions of its own. First, the dollar's value reflects expected future convenience yields and not only current trade flows and interest rates, so news alone can move the dollar (Jiang et al. 2021). Second, markets read a U.S. trade-policy shock as news about the future quality of dollar assets and not only as news about trade (Hassan et al. 2025). Under these assumptions the sign prediction of the safe-haven channel flips for exactly one type of event. Around the announcement of a shock that originates in U.S. policy, the erosion channel predicts

$$\text{CAR}(0, h) < 0 \quad \text{for short horizons } h$$

while the safe-haven channel alone would predict the opposite. Inside such an episode the dollar's usual positive relation to the VIX should also weaken or vanish, which the regression examines.

Gold separates the two channels. Its role as a crisis haven for equities exposes it to the safe-haven channel (Baur and Lucey 2010). At the same time it has no issuer, so no government's policy stands behind its value and the erosion channel cannot operate on it. Around a U.S.-originated trade-policy shock the two assets should therefore diverge. The dollar carries both channels and can fall, while gold carries only the safe-haven channel and should hold or rise. If the dollar loses its safe-haven response only when the shock is U.S.-originated while gold does not, the failure lies in the erosion channel and not in a general flight from safe assets.

3.3 Design of the Comparison

Because both channels respond to risk, a single episode cannot tell them apart. Separating them needs variation in two attributes of a shock, who originates it and what kind of shock it is. The two attributes have four combinations, and the three episodes studied fill three of them. The 2025 Liberation Day tariffs are a U.S.-originated trade-policy shock. The 2026 Israel–Iran/Hormuz crisis is a military/oil-supply shock in which the United States is involved as a co-belligerent. The 2022 invasion of Ukraine is an external military/oil-supply shock that the United States did not author. The fourth combination, an external trade-policy shock, has no counterpart in the sample. The twelve-day war of June 2025, which lifted crude only briefly, is reported alongside the Hormuz crisis as supporting context and stays outside the main formal cross-event comparison. Chapter 4 sets out the three episodes and the sources that document them. If only the safe-haven channel operates, all three episodes should appreciate the dollar and origin should not matter. If the erosion channel also operates, the U.S.-originated trade-policy shock should stand apart while the two military/oil-supply shocks resemble each other regardless of U.S. involvement. The two channels thus make opposite predictions about which episodes cluster together.

The design has limits. Each occupied combination holds a single episode. The two U.S.-linked shocks differ in the degree of involvement, as originator in one case and co-belligerent in the other. The fourth combination stays empty, so no purely external policy benchmark exists. The comparison therefore identifies patterns across these three episodes rather than properties of shock classes.

The oil market supplies both a classifier and a third channel. In the decomposition of [Kilian \(2009\)](#), the Hormuz episode is broadly supply-side, lifting crude through disrupted shipments and through precautionary demand about future supply, whereas the tariff shock depresses crude through weaker expected demand. The sign of the crude move therefore marks the nature of a shock. Through the commodity-currency channel the same move should also sort currencies, oil exporters outperforming importers when crude rises and underperforming when it falls ([Y.-C. Chen and Rogoff 2003](#); [H. Chen et al. 2016](#)).

The cross-section is read on the currency baskets of Chapter 5, the funding and carry portfolios ([Lustig et al. 2011](#)), the literature-standard baskets and the split into oil exporters and importers. Gold is read in dollars and in euros, so that mechanical denomination effects do not drive its measured response.

3.4 Hypotheses

The two channels lead to one overall prediction. The dollar's safe-haven response depends on the nature of a shock, not on U.S. involvement. Three ex-ante hypotheses make this testable.

The first hypothesis holds that responses cluster by the nature of a shock, not by U.S. involvement. Around the military/oil-supply shocks, Ukraine and Hormuz, the safe-haven channel governs and the dollar appreciates in both, whatever the degree of U.S. involvement. Around the U.S.-originated trade-policy shock the erosion channel governs and the dollar depreciates. The content of the first hypothesis is the joint pattern across the three episodes, not any single response.

The second hypothesis makes gold the origin-blind benchmark. Gold is exposed to the safe-haven channel but not to the erosion channel, so its response does not discriminate between shocks by origin. It holds or rises even in the U.S.-originated episode, in which the erosion channel predicts a weaker dollar. Any conditionality under the first hypothesis is then specific to the dollar.

The third hypothesis carries the oil channel into the cross-section. The exporter–importer spread follows the sign of the crude move. Oil exporters outperform importers when a shock lifts crude, as under Hormuz and Ukraine, and lose ground when it depresses crude, as on Liberation Day.

These predictions are not foregone, since [Yilmazkuday \(2025\)](#) finds the dollar depreciating in the year after global geopolitical-risk shocks. The first hypothesis fails if the tariff episode moves like the military pair or if the military pair splits along U.S. involvement. The second fails if gold weakens in the U.S.-originated episode. The third fails if the spread ignores the sign of the crude move. Chapter 6 reports the tests and Chapter 7 returns the verdicts.

4 Background

This chapter documents the episodes the empirical chapters analyse and establishes the dates that define the event windows of Chapter 5. For each episode it sets out what happened and how official institutions described the market reaction.

4.1 Liberation Day

On 2 April 2025, in an announcement branded “Liberation Day,” the United States imposed broad “reciprocal” tariffs on nearly all of its trading partners, combining a flat baseline rate of ten percent with higher country-specific surcharges. The measure was imposed under the International Emergency Economic Powers Act, after the President declared that large and persistent trade deficits and a lack of reciprocity in trade relationships amounted to a national emergency. It applied to most U.S. imports of goods ([Congressional Research Service 2026a](#)). Later assessments found that foreign exporters absorbed only about five percent of the tariff cost, so almost all of it fell on U.S. firms and consumers, while import volumes declined sharply ([European Central Bank 2026](#)).

Equity prices fell sharply and implied volatility spiked to a level exceeded only during the global financial crisis and the COVID-19 pandemic ([European Central Bank 2025](#)), while crude oil fell as markets priced slower growth and weaker oil demand ([U.S. Energy Information Administration 2026a](#)). Around the announcements the dollar depreciated against the other major advanced-economy currencies even as risk metrics deteriorated and Treasury yields rose ([Kamin 2025](#)). A partial ninety-day pause followed on 9 April 2025, suspending the country-specific surcharges while the baseline rate stayed in force ([Congressional Research Service 2026a](#)).

The legal basis of the tariffs remained contested. Multiple federal courts ruled that the President had exceeded his authority in at least some of the actions. The tariffs stayed in force while the Supreme Court reviewed the cases ([Congressional Research Service 2026a](#)). On 20 February 2026 the Supreme Court held that the act does not authorise the President to impose tariffs ([Supreme Court of the United States 2026](#)). Tariffs imposed under other trade statutes were not before the Court.

4.2 Twelve-Day War and Hormuz Crisis

Israel struck Iran on 13 June 2025, opening a twelve-day war that drew in U.S. strikes on three Iranian nuclear sites on 22 June. It ended on 24 June, hours after an Iranian missile strike on a U.S. base in Qatar ([Congressional Research Service 2025](#)). The episode raised the risk of a Gulf supply disruption and lifted crude only briefly ([Kilian et al. 2026](#); [U.S. Energy Information Administration 2026a](#)).

The conflict resumed on 28 February 2026, when the United States and Israel opened a joint campaign of airstrikes on Iran ([Congressional Research Service 2026b](#)). Both governments said they aimed to end Iran’s nuclear and missile capability and to force a change of government in Tehran. Iran’s response was regional, reaching Israel, U.S. military sites across the Middle East, and the Arab states that host American forces, among them Bahrain, Kuwait and Qatar. Iran also announced that it would close the Strait of Hormuz ([House of Commons Library 2026a](#)), which carries roughly a quarter of globally traded oil ([Fattouh and Economou 2026](#)). Attacks on shipping and the repricing of tanker insurance effectively closed the Strait within days ([Kilian et al. 2026](#)). On 13 March 2026 the United States reportedly struck military targets on Kharg Island, the terminal through which some ninety percent of Iran’s oil exports transit ([Congressional Research Service 2026b](#)). Brent crude, which had traded near \$71 per barrel before the 28 February strikes, rose by 46 percent over March to a monthly average of about \$104 and, in front-month futures, peaked at \$118 on 29 April ([Fattouh and Economou 2026](#); [International Monetary Fund 2026](#); [U.S. Energy Information Administration 2026c](#)). The daily spot series plotted in Figure 4.2 reached \$138.21 on 7 April ([U.S. Energy Information Administration 2026b](#)).

A two-week ceasefire announced on 7 April 2026 came with an Iranian declaration that the Strait was open, though only to vessels using a route coordinated with Iranian port authorities, which falls short of free passage ([Fattouh and Economou 2026](#)). Talks in Islamabad on 11 and 12 April broke down, and the United States began a naval blockade of Iranian ports on 13 April. The ceasefire itself held into May, but Iran resumed attacks on 4 May and the United States struck Iran again on 25 May ([Congressional Research Service 2026c](#); [CNN 2026](#)). A memorandum of understanding signed on 17 June set out plans for talks over an initial sixty-day period on freedom of navigation, Iran’s nuclear and missile programmes and sanctions ([House of Commons Library 2026b](#)). Once it was signed and tanker movements picked up, Brent fell through the rest of the quarter to \$72 on 26 June ([U.S. Energy Information Administration 2026c](#)).

The fighting did not stop with the agreement. Between 25 and 28 June the United States struck Iran in response to an attack on commercial shipping in the Strait, and Iran struck U.S. bases in Bahrain and Kuwait ([House of Commons Library 2026b](#)). The sample analysed in this thesis ends on 30 June 2026, with the conflict unresolved. Israel's part in the fighting was concentrated in the first weeks. An exchange of fire with Iran on 8 June was its first since the April ceasefire, and both sides then backed away from further strikes ([Gambrell and Lidman 2026](#)). Israel was a partner in the military campaign but had no part in shaping the memorandum of 17 June ([Shavit and Shine 2026](#)), and it sat out the fighting between the United States and Iran in July ([Berman 2026](#)).

4.3 Russian Invasion of Ukraine

On 24 February 2022 Russian forces crossed the border into Ukraine and began a full-scale assault on the country. The European Union, the United Kingdom, the United States and other allies answered within days with coordinated sanctions targeting Russia's banks, its aviation and shipping sectors, and industries of strategic significance to the Russian state, among them defence, aerospace and energy ([House of Commons Library 2025](#)).

Two measures reached the international payments and reserve system directly. On 26 February 2022 the European Union, the United Kingdom, the United States and Canada announced that they would restrict the Russian Central Bank's access to its foreign reserves. The stated aim was to undercut its ability to conduct foreign exchange transactions in support of the rouble. Russia was thought to hold more than \$630 billion in currency and gold reserves at the time. By late 2023 more than sixty percent of them had been frozen. On 2 March 2022 the European Union announced the exclusion of seven Russian banks from the SWIFT (Society for Worldwide Interbank Financial Telecommunication) payments network, effective ten days later. Three more followed in June ([House of Commons Library 2025](#)).

Energy markets had already been pricing the risk of an invasion. Crude oil rose and grew more volatile from mid-January 2022 onward, passed \$100 per barrel as the attack began, and Brent futures closed just below \$115 per barrel on 2 March ([U.S. Energy Information Administration 2022](#)).

4.4 Overview of the Events

The analysis dates six events. The two treatments are Liberation Day, a trade-policy shock the United States authored, on 2 April 2025, and the Hormuz crisis, a military and oil-supply shock in which it was a belligerent, on 28 February 2026. Each is followed by a reversal within weeks, the ninety-day tariff pause on 9 April 2025 and the two-week ceasefire on 7 April 2026. The invasion of Ukraine on 24 February 2022 enters as the external control, a shock the United States did not author, and the twelve-day war of 13 June 2025 enters as supporting context, reported alongside the Hormuz crisis but outside the main formal cross-event comparison.

Figures 4.1–4.3 place these events in their longer-run context, plotting the broad dollar index, the price of Brent crude, and the geopolitical-risk index of [Caldara and Iacoviello \(2022\)](#) over 2019–2026, with the four main events marked on each series. Chapter 6 measures the responses formally.

4 Background



Figure 4.1: Broad dollar index, 2019–2026. Source: [Board of Governors of the Federal Reserve System \(2026c\)](#), via FRED (Federal Reserve Economic Data).



Figure 4.2: Brent crude oil price, 2019–2026, USD per barrel. Source: [U.S. Energy Information Administration \(2026b\)](#), via FRED.

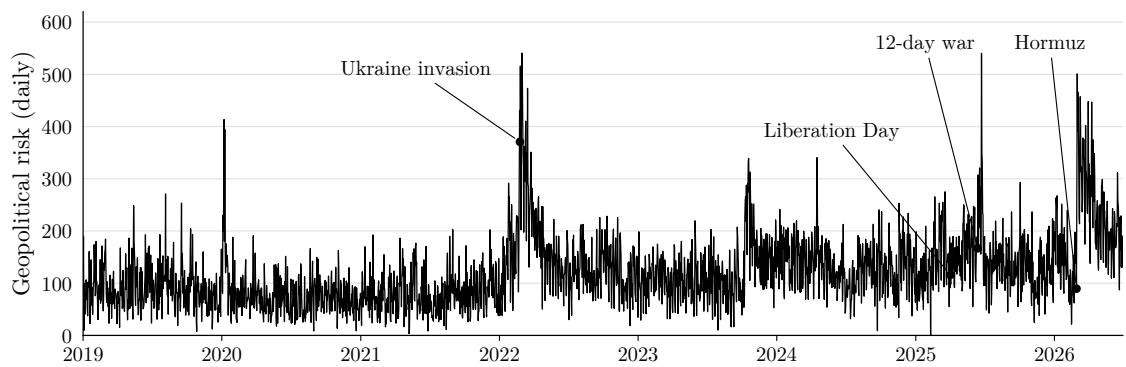


Figure 4.3: Daily Geopolitical Risk Index, 2019–2026. Source: [Caldara and Iacoviello \(2026\)](#).

5 Data and Methodology

This chapter describes the data and the empirical methods used for the analysis in Chapter 6. Section 5.1 describes the exchange rates and financial variables, the geopolitical and policy uncertainty indices, and the construction of the currency baskets. Section 5.2 sets out the event study and Section 5.3 the regression.

5.1 Data

5.1.1 Exchange Rates and Financial Variables

The currency panel contains daily exchange rates for 32 currencies against the U.S. dollar from 1 January 2019 to 30 June 2026. Spot rates are the mid of the closing bid and ask quotes from LSEG Workspace ([LSEG 2026d](#)), with the Federal Reserve’s H.10 release as a cross-check ([Board of Governors of the Federal Reserve System 2026a](#)). All rates are expressed as foreign currency per dollar, and the four that LSEG quotes as dollars per currency (EUR, GBP, AUD, NZD) are inverted. Daily returns are computed as log changes,

$$r_{i,t} = \ln(S_{i,t-1}/S_{i,t}),$$

where $S_{i,t}$ is the spot rate of currency i on day t . A positive return is an appreciation of the foreign currency against the dollar.

The remaining series come from FRED and LSEG. FRED provides the nominal broad dollar index, the 10-year Treasury yield, the VIX, the Brent and West Texas Intermediate (WTI) crude prices and the S&P 500 index ([Board of Governors of the Federal Reserve System 2026c](#); [Board of Governors of the Federal Reserve System 2026b](#); [Cboe Global Markets 2026](#); [U.S. Energy Information Administration 2026b](#); [S&P Dow Jones Indices 2026](#)). Brent serves as the global crude benchmark and WTI as the U.S. benchmark ([U.S. Energy Information Administration 2022](#)). LSEG provides the gold price in dollars and the Euro Stoxx 50 index in euros ([LSEG 2026a](#)). Series identifiers are listed in Chapter 10. Indices, equities and commodities enter as log returns. The 10-year yield and the VIX enter in first differences.¹

¹Missing prices are filled with the previous quote for up to three consecutive weekdays. Re-estimating means and standard deviations after excluding zero returns from the original estimation windows changes the broad-dollar Liberation Day–Hormuz CAR(0, +20) comparison from $t = -4.11$ to -4.03 , leaving significance at the 1 percent level unchanged.

Because oil and gold are priced in dollars, part of any change in their dollar price reflects the dollar itself rather than the commodity. A depreciation of the dollar raises the dollar price of oil even when oil is unchanged in every other currency. To separate commodity movements from this denomination effect, Brent and gold are additionally re-expressed in euros as

$$r_{X,t}^{\text{EUR}} = r_{X,t}^{\text{USD}} - r_{\text{EUR},t}, \quad (5.1)$$

where X denotes Brent or gold and $r_{\text{EUR},t}$ is the euro's log appreciation against the dollar. The euro-denominated series enter the event study as a robustness check on denomination.

5.1.2 Geopolitical and Policy Uncertainty Indices

Geopolitical risk is measured by the daily Geopolitical Risk (GPR) index of [Caldara and Iacoviello \(2022\)](#) and trade-policy uncertainty by the daily Trade Policy Uncertainty (TPU) index of [Caldara et al. \(2020\)](#), both downloaded from the authors' site ([Caldara and Iacoviello 2026](#); [Caldara et al. 2026](#)).² Both indices count newspaper coverage. The GPR index measures the share of articles discussing adverse geopolitical events in ten leading U.S., British and Canadian newspapers, and the TPU index the share of articles discussing trade-policy uncertainty in seven major newspapers. The J.P. Morgan VXY Global index of implied U.S.-dollar exchange-rate volatility is the currency-market counterpart of the VIX ([LSEG 2026b](#)). The one-month forward discount $f_{i,t} - s_{i,t}$, the log forward minus the log spot rate, with the forward rates from LSEG Workspace ([LSEG 2026c](#)), proxies the interest-rate differential under covered interest parity ([Lustig et al. 2011](#)) and is the sorting variable for the currency baskets of Section 5.1.3.

5.1.3 Portfolio Construction

The safe-haven prediction concerns types of currencies rather than any single one, so the currencies enter the event study in baskets. The central pair, a safe and a risky basket, is defined in two ways. In the first, currencies are sorted into terciles by their average one-month forward discount over the sample. The low-yield tercile forms the safe basket and the high-yield tercile the risky basket, with pegged and

²The TPU series is subject to revision. The analysis uses the version downloaded on 23 July 2026.

managed currencies (DKK, HKD, SAR, KWD, CNY) excluded from the sort.³ Four further currencies (BRL, CLP, COP, PEN) trade forward only as non-deliverable contracts without a clean one-month quote and drop from the currency sort for lack of data, leaving 23 currencies in the sort. The choice of terciles follows the sizing convention of the sorting literature, which builds fewer, broader portfolios when there are few assets to sort. [Asness et al. \(2013\)](#) sort the ten currencies in their sample into three equal groups and [Lustig et al. \(2011\)](#) form six portfolios from at most 35 currencies. Sorting the 23 currencies into terciles gives the safe and the risky basket seven each. Quintiles would leave four or five, few enough that a single currency's swings could dominate the basket. This fixed tercile classification defines the safe and risky currency baskets used in the main event-study specification (the full classification is given in Table 1 in the Appendix). Because the low-yield tercile also contains low-rate Asian currencies that are not conventional safe havens, a second, literature-standard scheme is reported as a robustness check, pairing the yen, Swiss franc and euro as the safe basket ([Ranaldo and Söderlind 2010](#)) with the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah as the risky basket ([De Bock and de Carvalho Filho 2015](#)). The euro enters the literature-standard safe basket as the largest liquid alternative to the dollar rather than as a documented safe haven. Direct haven evidence for the euro is confined to the period before 2008 ([Habib and Stracca 2012](#)), but recent work on the dollar's reserve status treats the euro as the principal alternative currency ([Jiang et al. 2026](#)). A separate split tracks the oil channel. Currencies are grouped by their observed net crude oil trade over 2019–2024 into oil exporters (NOK, CAD, MXN, COP, BRL, SAR, KWD) and oil importers (JPY, KRW, INR, THB, TWD, TRY) ([United Nations Statistics Division 2026](#)).⁴ [H. Chen et al. \(2016\)](#) apply the same split to the six of these currencies in their sample, grouping Canada, Norway and Mexico as oil exporters and Japan, Korea and Turkey among the importers. A higher oil price improves the exporters' terms of trade and worsens the importers', so the two baskets should separate in oil-driven shocks ([Y.-C. Chen and Rogoff 2003](#); [H. Chen et al. 2016](#)). All baskets are equally weighted, the convention of the sorting literature. The dollar basket averages the dollar's return against all 32 currencies.

³HKD and SAR are pegged to the dollar, KWD is linked to a currency basket and DKK to the euro, while CNY is managed ([International Monetary Fund 2025a](#)). DKK is excluded as a euro duplicate (sample correlation of daily returns against the dollar 0.998). The other four regimes constrain exchange-rate adjustment.

⁴Partner reports supplement missing reporter observations. For Türkiye, imports are measured by partners' exports to Türkiye and exports by partners' imports from Türkiye. Table 1 states the remaining coverage gaps.

5.2 Event Study Methodology

The main specification is the constant-mean return model ([MacKinlay 1997](#); [Campbell et al. 1997](#)). The event window is $[-20, +20]$ trading days. For each series i the normal return is its average over an estimation window of $[-140, -21]$ trading days preceding the event, ending where the event window begins so that no event-window day enters the estimation of the benchmark. The abnormal return on day t of the event window is the deviation from that mean,

$$AR_{i,t} = R_{i,t} - \hat{\mu}_i, \quad \hat{\mu}_i = \frac{1}{120} \sum_{s=-140}^{-21} R_{i,s}, \quad (5.2)$$

and the cumulative abnormal return aggregates these over the window,

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}.$$

Each event is dated by its announcement, the point at which the information reaches markets, and that trading day is day 0. When an announcement falls on a non-trading day, day 0 is the first trading day after it, so the strikes of 28 February 2026 enter on Monday 2 March 2026. Because daily prices react to many things at once, the longer the window over which abnormal returns are summed, the weaker the claim that the measured movement belongs to the event alone ([MacKinlay 1997](#); [Kothari and Warner 2007](#)). Inference therefore rests on the short windows. The symmetric $CAR(-1, +1)$ guards against information reaching prices the day before the announcement, and $CAR(0, +1)$ and $CAR(0, +5)$ capture the announcement reaction. Both tariff announcements, Liberation Day and the tariff pause, came in the U.S. afternoon, just after 4 p.m. Eastern time on 2 April 2025 ([Benguria and Saffie 2025](#)) and in the early afternoon of 9 April 2025. That is after the noon pricing of the H.10 dollar indices and late in the pricing day of the currency panel, so the day-0 return contains at most a first partial reaction and the full response appears on the following trading day. Capturing announcements that arrive around or after the close is the standard reason for the two-day window. The longer cumulations $CAR(0, +10)$ and $CAR(0, +20)$ are reported as robustness checks on the durability of the response rather than as independent identification, and a long horizon $CAR(0, +50)$, a persistence check, is estimated on $[-170, -51]$ so the estimation window stays clear of this check's wider $[-50, +50]$ event window.

The significance of a single CAR is assessed with

$$t = \frac{\text{CAR}_i(t_1, t_2)}{\hat{\sigma}_i \sqrt{L}}, \quad L = t_2 - t_1 + 1, \quad (5.3)$$

where $\hat{\sigma}_i$ is the estimation-window standard deviation of $\text{AR}_{i,t}$. The statistic is compared with the standard normal distribution, and the tables mark significance at the 10, 5 and 1 percent levels with its critical values of 1.65, 1.96 and 2.58. The difference between two events' CARs is tested with

$$\Delta = \text{CAR}_A(t_1, t_2) - \text{CAR}_B(t_1, t_2), \quad \text{s.e.}(\Delta) = \sqrt{L(\hat{\sigma}_A^2 + \hat{\sigma}_B^2)}, \quad (5.4)$$

where $\text{s.e.}(\Delta)$ is the standard error of the difference, the two-sample analogue of the CAR variance in [Campbell et al. \(1997\)](#). The two events are treated as independent because their windows do not overlap.

The constant-mean model is preferred to the market model because the object of study is the dollar itself. The choice matters in currency applications. [Kwok and Brooks \(1990\)](#), comparing competing abnormal-return specifications in foreign-exchange markets, find inference sensitive both to the currency in which returns are measured and to the benchmark model, so event-study practice developed for equities does not carry over automatically. When an event moves many currencies at once, a market model subtracts the common movement, the very thing this study wants to measure. Kwok and Brooks therefore prefer mean-adjusted returns when the whole unexpected movement is the quantity being measured. The natural market return for the currency baskets is the equal-weighted dollar basket, the average of the panel itself. Regressing dollar exchange rates on it would absorb the common dollar movement into the market term and leave little in the abnormal return. The market model is nevertheless reported as a robustness check,

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \quad \text{AR}_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}), \quad (5.5)$$

where $R_{m,t}$ is the equal-weighted dollar basket. The basket-level results in the Appendix leave the safe-haven ordering under both treatment shocks unchanged. When many correlated securities share a common event date, cross-sectional dependence inflates the test statistics, and the adjusted test of [Kolari and Pynnönen \(2010\)](#) corrects for it. Neither condition applies here. Inference is drawn at the basket level around single, non-overlapping event dates, so the plain t -test is kept.

5.3 Regression Specification

The event study measures whether the dollar moved. The regression measures whether its relationship with risk changed. The classical safe-haven property is that the dollar appreciates when global risk rises (Habib and Stracca 2012), so the daily broad-dollar return is regressed on a daily risk shock ΔR_t , interacted with a dummy for each crisis window, following the interaction tests of Baur and Lucey (2010),

$$r_t^{\text{USD}} = a + b_1 \Delta R_t + \sum_k \beta_k (\Delta R_t \times D_t^k) + \sum_k \gamma_k D_t^k + \varepsilon_t, \quad (5.6)$$

where the sum runs over the three crisis windows, $k \in \{\text{tariff, Hormuz, Ukraine}\}$. The dummy D_t^k marks the window of shock k . It is one on the event day and the twenty trading days after it, and zero on all other days. The coefficient b_1 gives the dollar's response to a risk shock on days outside these windows. The β_k measure how that response changes inside each crisis window, and the γ_k capture the dollar's average drift within each window, so a window in which the dollar simply trended does not show up as a changed risk response. The event dates and the twenty-day horizon are the same as in the event study, so both methods work with the same definition of each episode.⁵ The equation is estimated by ordinary least squares (OLS) with Newey–West standard errors with five lags (Newey and West 1987), which are heteroskedasticity- and autocorrelation-consistent (HAC) and therefore robust to the volatility clustering of daily returns. All computations use Python (version 3.14) with the pandas, statsmodels and matplotlib libraries, and the code and data are provided together with the thesis.

The choice of ΔR_t matters because no single index captures both shock types equally well. Over the two-day window (0, +1), the VIX responds strongly to the trade-policy shock but only mildly to the Gulf escalation, and the news-based GPR index does the opposite and barely registers a tariff announcement. The regression is therefore estimated in four versions. The baseline uses the change in the VIX. The second adds the GPR index alongside the VIX, each with its own interaction terms, so stock-market risk and geopolitical risk enter separately. Their daily changes are almost uncorrelated, $\rho = -0.04$, so the data can tell the two effects apart. The third replaces both with the VXY Global index of implied U.S.-dollar exchange-rate volatility, a measure taken from the currency market itself. The fourth combines

⁵The dynamic conditional correlation model of Engle (2002) allows conditional correlations to vary over time. The interaction specification tests changes in the dollar's sensitivity to risk within predefined crisis windows.

5 Data and Methodology

the two news-based indices, GPR and TPU. Each risk shock is scaled by its own standard deviation over the full sample, so the coefficients are comparable across versions and read as the percent dollar response to a one-standard-deviation rise in risk. Results are reported in Section 6.6.

6 Empirical Results

This chapter presents the thesis’s empirical results. An event study, set out in detail in the next five sections, measures how the U.S. dollar, the cross-section of currency baskets, crude oil, gold and equities responded to the four shocks, the Liberation Day tariffs of April 2025, the twelve-day war of June 2025, the Hormuz crisis of 2026 and the Russian invasion of Ukraine in 2022. The regression in Section 6.6 then tests whether the relationship with risk that makes the dollar a safe haven held or broke across the episodes. Its results are read as supporting evidence for the event study rather than as an independent test.

Three of the four shocks are U.S.-linked. The Liberation Day tariff announcement is U.S.-originated, and the twelve-day war and the Hormuz crisis are military shocks in which the United States is involved as a co-belligerent. The Russian invasion of Ukraine is an external military/oil-supply shock that the United States did not author, and thus serves as the benchmark for the “classical” safe-haven response. The formal cross-event comparison of Section 6.3 rests on Liberation Day, the Hormuz crisis and the Ukraine benchmark. The twelve-day war is reported as supporting context and stays outside that comparison.

Abnormal returns are computed with the constant-mean model of Section 5.2. The normal return of each series is its mean over the estimation window of $[-140, -21]$ trading days, the abnormal return is the deviation from that mean, and the cumulative abnormal return $CAR(t_1, t_2)$ sums the abnormal returns over the event window from day t_1 to day t_2 . Conclusions are drawn from the short windows $CAR(-1, +1)$, $CAR(0, +1)$ and $CAR(0, +5)$, which capture the reaction to the announcement itself. Over longer windows other events affect returns as well, so the ten-, twenty- and fifty-day CARs are used to trace how long a response lasts, not to establish that there was one. Significance is assessed with the t -test in equation (5.3). Tables 6.1 and 6.2 collect the main results, and the discussion below reads them event by event. The figures plot Brent, the global benchmark, and WTI is reported in Table 6.2.

6.1 Dollar Response to the Tariff Shock

The tariff announcement of 2 April 2025 triggered an immediate depreciation of the dollar. The broad dollar index fell 1.44% over the two-day announcement window and 1.70% over the symmetric $(-1, +1)$ window, both significant at the 1% level (Table 6.1, Panel A). The five-day CAR of -0.10% is close to zero because the dollar

6 Empirical Results

recovered on 4 and 7 April, undoing the initial fall within the window. Day +5 falls on 9 April 2025, the day of the ninety-day tariff pause, so the five-day window ends on a second policy event. $CAR(-1, +1)$ and $CAR(0, +1)$ end on 3 April, four trading days before the pause, which makes them the cleaner measures of the announcement effect. The pause is dated as a sub-event with 9 April as day 0. Over its two-day window, 9 and 10 April, the dollar falls again, by 1.48%, and the fall continues, to -3.07% after five trading days and -4.51% after twenty. Measured from the announcement, the move persists rather than reverses over the longer horizons, -3.00% by ten trading days, -3.98% by twenty and -6.34% by fifty (Section 6.5), which is consistent with markets reading the tariffs as a lasting change in U.S. trade policy rather than a one-off shock.

U.S. markets responded with a broad flight from risk. Over the two-day announcement window, 2 and 3 April, the S&P 500 fell 4.33% and the VIX rose 8.15 points, both significant at the 1% level, while the 10-year Treasury yield fell 11.95 basis points, a move that is not statistically significant. Equities sold off, Treasuries did not, and the dollar fell alongside equities rather than rising against them. Around the pause Treasuries sold off as well, the 10-year yield rising 23.65 basis points over the three days from 8 to 10 April, significant at the 5% level, while the dollar kept falling. Over 2 and 3 April the yen, Swiss franc and euro rose against the dollar by 2.48%, 2.89% and 2.44% respectively. Together they form the literature-standard safe basket, which rose 2.60%. The broader carry-sorted safe basket, which also contains low-yield Asian currencies, rose 1.52%. All of these moves are significant at the 1% level. Over five days the gains shrink to 2.12% for the literature-standard safe basket and 1.07% for the carry-sorted basket, because the dollar's recovery on 4 and 7 April pulled both back (Table 6.2). These five-day gains are significant only at the 10% level for the literature-standard safe basket, while not significant for the carry-sorted basket.

The oil-exporter and oil-importer baskets point the same way. Crude fell sharply on the announcement (Section 6.4), and over ten days the importer basket gained 2.84% while the exporter basket was essentially flat at +0.04%, a gap of 2.80 percentage points, significant at the 1% level (Figure 6.1). The gap does not isolate the oil channel, however. Three of the six importer currencies, the yen, baht and Taiwan dollar, also belong to the carry-sorted safe basket and had already risen as safe currencies. Taken together, these results match what the framework calls the erosion channel. A U.S.-originated shock weakened the dollar, and the demand for safety went to the yen, franc and euro instead, while oil and equities fell.

6 Empirical Results

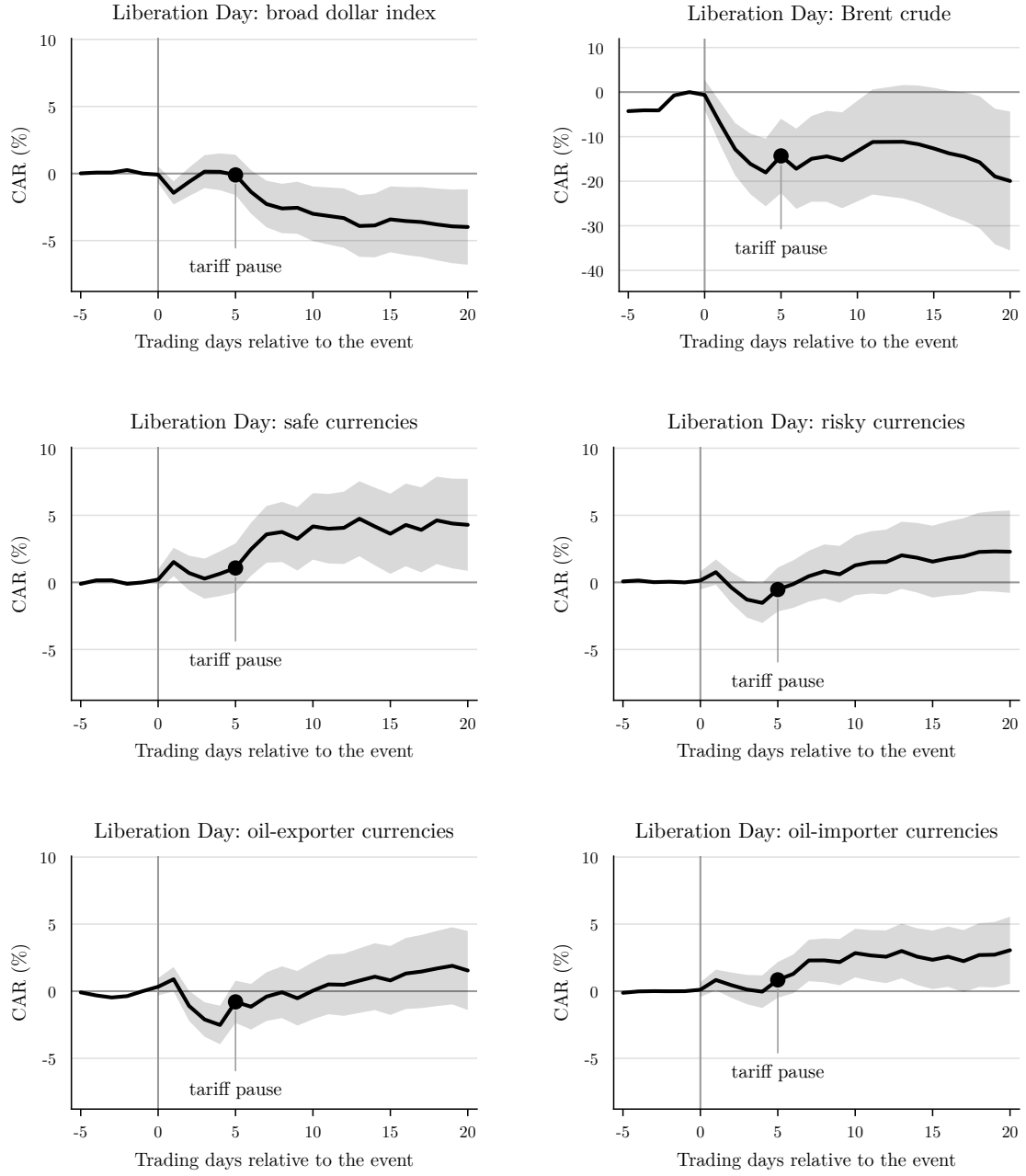


Figure 6.1: Liberation Day (2 April 2025): cumulative abnormal returns over -5 to $+20$ trading days, constructed as in Section 5.2. In each portfolio figure the dollar-index and currency-basket panels share one vertical scale, chosen for that figure, and Brent crude has its own. Every path is re-anchored to zero at day -1 so that the movement from day 0 onwards is the event response. The shaded band is ± 1.96 standard errors of the cumulative abnormal return as defined in Section 5.2. Where it clears zero, $CAR(0, h)$ is significant at the 5% level. Dots mark dated sub-events.

6 Empirical Results

Table 6.1: The U.S. dollar's response by event and horizon

Event	Headline windows			Longer horizons		
	(−1, +1)	(0, +1)	(0, +5)	(0, +10)	(0, +20)	(0, +50)
<i>Panel A. U.S.-originated trade-policy shock</i>						
Liberation Day	−1.70***	−1.44***	−0.10	−3.00***	−3.98***	−6.34***
Tariff pause	−1.48**	−1.48***	−3.07***	−3.44***	−4.51***	−6.74***
<i>Panel B. U.S.-involved military/oil-supply shocks</i>						
Twelve-day war	−0.28	+0.04	+0.88	−0.13	+0.35	−0.12
Hormuz crisis	+1.34***	+1.39***	+1.53***	+2.10***	+3.25***	−0.13
Hormuz ceasefire	−1.33***	−1.14***	−1.72***	−1.65**	−1.47	−0.04
<i>Panel C. External control</i>						
Ukraine invasion	+0.45	+0.49	+1.03*	+1.65**	+0.84	+3.12**

Notes. Cumulative abnormal returns of the broad nominal U.S. dollar index, in percent, constant-mean model. Estimation windows are as in Section 5.2, with the fifty-day column estimated on $[-170, -51]$. The three headline windows carry the conclusions. The longer horizons trace how long a response lasts (Section 6.5). Positive values denote dollar appreciation. The tariff pause and the Hormuz ceasefire are sub-events within their parent windows. The ceasefire's estimation window ends on 9 March and contains the first six trading days of the crisis, so its cells are read with corresponding caution. The May 2026 U.S. strikes and the June 2026 memorandum are not analysed (Section 6.5). *, **, *** denote significance at the 10%, 5% and 1% level.

6.2 Dollar Response to the Military Shocks

The dollar responded very differently to the two military shocks. It showed no significant move in the twelve-day war and rose as a classical safe haven in the Hormuz crisis. The twelve-day war lifted crude only briefly, and the United States joined it six trading days after Israel's opening strike. The Hormuz crisis disrupted oil supply for months, and the United States was a belligerent from the first day.

The twelve-day war of June 2025 produced no statistically significant dollar response. The broad dollar index moved +0.04% over the two-day window and +0.88% over five days, neither significant (Table 6.1, Panel B), while Brent rose 11.25% over the same five days, significant at the 1% level (Table 6.2). The full set of portfolio panels for this episode is shown in the Appendix. The U.S. strikes of 22 June 2025 enter the panel on day +6, so the three headline windows cover an opening week in which the war was fought between Israel and Iran alone, and only the ten- and twenty-day windows contain the U.S. strikes and the ceasefire of 24 June. The dollar moved significantly in neither stretch. Up to day +6 its cumulative abnormal return

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drifted to +1.01%, the same sign as the Ukraine benchmark but never significant, and by day +9 the gain was gone. An offset between the flight-to-the-dollar reflex and the erosion pressure once the United States took part would fit that reversal, but the ceasefire on the next trading day fits it equally well, so the episode identifies neither. The response is read as descriptive, and Section 6.5 adds that its benchmark is distorted by the Liberation Day crash, which leaves even its sign uncertain.

In the Hormuz crisis the dollar behaved as a classical safe haven. The broad index rose 1.39% over the two-day window and 1.53% over five days, both significant at the 1% level, and kept rising to +2.10% at ten days and +3.25% at twenty (Table 6.1, Panel B). Crude rose sharply from the first day (Section 6.4), but unlike under the tariff shock the oil price did not sort the currencies. Both oil baskets fell against the rising dollar over five days, the importers by 1.03% and the exporters by 0.69%, and the exporter-minus-importer spread was not significant at any horizon, +0.54 percentage points at twenty days (Figure 6.2). The sign of the gap fits higher crude cushioning the exporters, but the difference never became significant. The dollar rose even though the United States was a belligerent from the first day, so for a military/oil-supply shock the safe-haven reflex survived U.S. involvement, in contrast to the trade-policy shock. U.S. assets show the same contrast. The S&P 500 was little changed over the first week, and the 10-year Treasury yield rose 26.64 basis points by ten days and 39.22 by twenty, both significant at the 5% level, so yields moved with oil rather than with a flight into Treasuries while the dollar appreciated.

The Hormuz appreciation was temporary. By fifty days the broad-index CAR had returned to an insignificant -0.13% , so most of the move unwound. The clearest evidence that this was a genuine but temporary risk premium comes from the ceasefire of 7 April 2026, a sub-event within the crisis marked at day +26 in Figure 6.4. On the two-week ceasefire announcement, which came with an Iranian declaration that the Strait was open again to coordinated traffic, the dollar gave back 1.14% over the two-day window and 1.72% over five days, both significant at the 1% level, and Brent fell 21.58% within ten days. The premium that the threat put into the dollar came back out on de-escalation, which ties the exchange-rate move to the geopolitical risk it priced. The ceasefire's estimation window ends on 9 March and therefore contains the first six trading days of the crisis, so its numbers are read with the caution noted in Table 6.1.

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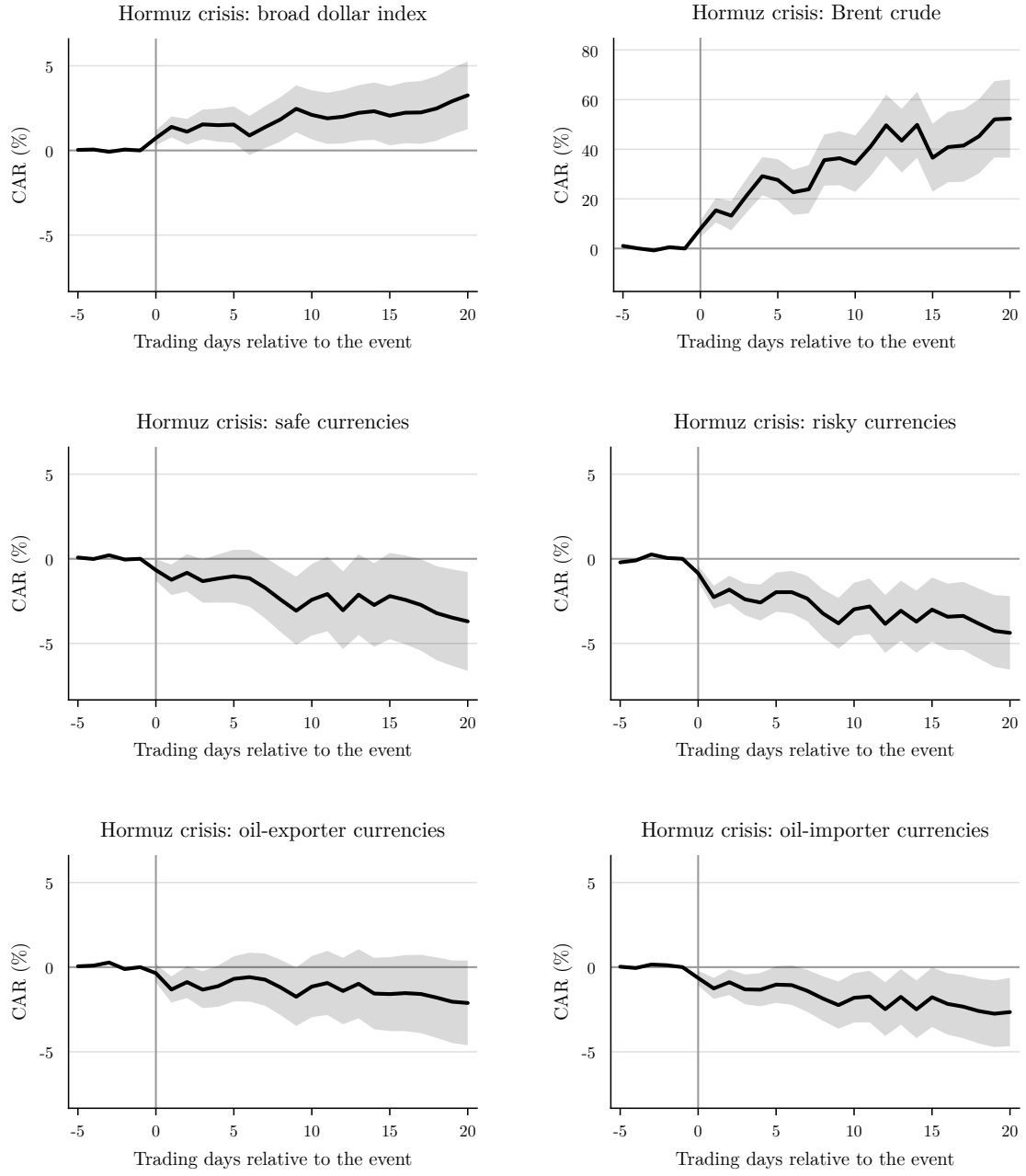


Figure 6.2: Hormuz crisis (28 February 2026): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1.

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Table 6.2: Cross-section of five-day responses: the mirror image

Series, CAR(0, 5) (%)	U.S.-originated	U.S.-involved		Control
	Liberation Day	Hormuz	Twelve-day war	Ukraine
Broad USD index	−0.10	+1.53***	+0.88	+1.03*
Dollar (equal-weighted)	+0.21	+1.36**	+0.83	+1.01
Safe currencies (carry)	+1.07	−1.03	−1.04	−1.07*
Risky currencies (carry)	−0.53	−1.97***	−1.03	−2.56*
Oil exporters	−0.80	−0.69	−0.45	+0.30
Oil importers	+0.84	−1.03*	−1.12	−0.74
Gold	−1.54	−4.54	−1.53	+1.25
S&P 500	−3.30*	−1.58	−1.22	+3.32
Euro Stoxx 50	−14.58***	−8.11***	−3.05	−5.96**
Brent crude	−14.32***	+27.63***	+11.25***	+13.82***
WTI crude	−13.14***	+34.55***	+10.14**	+14.38***
<i>Robustness: literature-standard baskets</i>				
Safe currencies (literature)	+2.12*	−1.23	−1.33	−0.69
Risky currencies (literature)	−1.69*	−1.18	−0.36	+0.24
<i>Robustness: euro-denominated commodities</i>				
Brent crude in EUR	−15.99***	+29.21***	+12.09**	+15.79***
Gold in EUR	−3.21	−2.97	−0.70	+3.23*

Notes. Five-day cumulative abnormal returns (%), constant-mean model, estimation window $[-140, -21]$. Positive values are appreciation against the dollar for the currency baskets and a rise in the index or price for all other series. *Safe/Risky (carry)* are the bottom and top forward-discount terciles (Section 5.1.3). The literature-standard baskets are listed in the Appendix. The bottom panel re-expresses the commodities in euros (Section 5.1.1). Gold's CARs are read for sign rather than magnitude (Section 6.4). *, **, *** denote significance at the 10%, 5% and 1% level.

6.3 Cross-Event Comparison

Reading across Tables 6.1 and 6.2, the two treatment shocks produce a mirror image. The trade-policy shock weakens the dollar and depresses oil, while the military/oil-supply shock strengthens the dollar and lifts oil. The dollar is therefore not a fixed safe haven or a fixed casualty. Its response is conditioned by the type of shock, even holding U.S. involvement fixed.

This contrast is not merely descriptive. Table 6.3 tests the difference in the dollar's CAR between events directly, and the separation is immediate. Already over the first two trading days the broad dollar index is 2.83 percentage points lower under the tariff shock than under the Hormuz crisis ($t = -5.21$) and 1.93 points lower than under Ukraine ($t = -3.53$), both significant at the 1% level. The five-day difference narrows to −1.63 points ($t = -1.73$) as the dollar's recovery of 4 and 7 April, just

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before the pause, temporarily retraces its fall, before the gap re-opens to 5.10 points at ten days ($t = -4.00$) and 7.24 points at twenty ($t = -4.11$). Against Ukraine the pattern is the same (-4.65 at ten days, $t = -3.63$). The Hormuz and Ukraine responses, by contrast, share the same sign throughout. Over the first two days the Hormuz bid is somewhat stronger ($+0.91$ points, $t = 2.02$), a difference of intensity rather than direction, at five and ten days the two are statistically indistinguishable, and at twenty days a marginal gap re-opens ($+2.41$, $t = 1.66$). The mirror image is thus driven by the trade-policy shock, the only episode in which the dollar moves against the classical safe-haven direction, and it does so within the first two trading days.

Table 6.3: Difference in dollar CARs across events (broad USD index, %)

Contrast (A – B)	$\Delta\text{CAR}(0, h)$, percentage points			
	$h=1$	$h=5$	$h=10$	$h=20$
Liberation Day – Hormuz crisis	-2.83^{***}	-1.63^*	-5.10^{***}	-7.24^{***}
Liberation Day – Ukraine	-1.93^{***}	-1.14	-4.65^{***}	-4.83^{***}
Hormuz crisis – Ukraine	$+0.91^{**}$	$+0.50$	$+0.45$	$+2.41^*$

Notes. Difference between two events' cumulative abnormal returns of the broad nominal U.S. dollar index (%), constant-mean model, estimation window $[-140, -21]$. Standard errors follow equation (5.4). Each window includes day 0 through day h , and events are treated as independent because their windows do not overlap. The twelve-day war is not paired with Liberation Day because its estimation window contains the Liberation Day event window (Section 6.5). A positive value means the dollar rose more (or fell less) under shock A than shock B. */**/***: 10/5/1%.

The Ukraine benchmark disciplines the interpretation. As an external military/oil-supply shock, it generates the textbook response. The dollar appreciates ($+1.65\%$ at ten days, $+3.12\%$ at fifty, Table 6.1) and crude rises 13.82% (Table 6.2). One caveat concerns the dating of this benchmark. The invasion was partly anticipated. Energy markets had priced the risk since mid-January (Section 4.3), and the VIX rose from 19.96 on 9 February 2022 to 31.02 on 23 February, the last trading day before the invasion, then eased to 27.59 on day $+1$. The dollar shows no such run-up, with a cumulative abnormal return of -0.63% ($t = -0.63$) over the twenty trading days before the invasion and -0.09% over the last five, and its appreciation begins on day 0 with an abnormal return of $+1.09\%$ on the invasion day itself. The Ukraine CARs are therefore read as the response to the invasion rather than to the weeks before it, and the flat path before day 0 means that the event window does not miss an earlier dollar move. As just shown, that response shares its sign throughout the first twenty trading days and, at five and ten days, its magnitude with Hormuz. Two features nonetheless distinguish the external from the U.S.-involved military

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shock. First, the Ukraine appreciation is durable whereas the Hormuz appreciation reverts, consistent with an external shock conferring a lasting safe-haven bid and a U.S.-involved one only a temporary one. Second, and most importantly for the thesis, the trade-policy shock breaks the pattern entirely. The only episode in which the dollar depreciates is the one the United States itself authored on the economic-policy front. The safe-haven function is intact for geopolitical/oil-supply shocks, including U.S.-involved ones, but it inverts for a U.S. trade-policy shock.

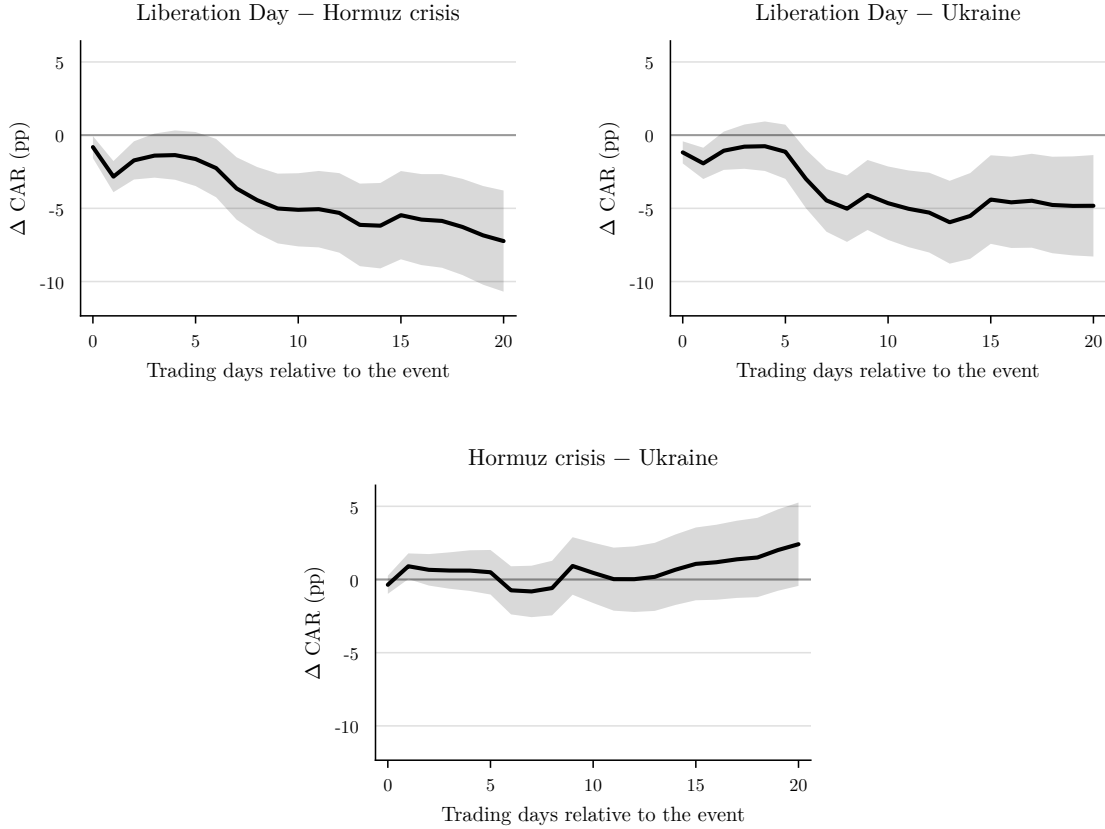


Figure 6.3: Difference in the broad dollar index's cumulative abnormal return between events, in percentage points over the first twenty trading days. The differences and their standard errors follow equation (5.4). The three contrasts are those of Table 6.3. All three panels share one vertical scale. The shaded band is ± 1.96 standard errors of the difference. Where it clears zero, the difference is significant at the 5% level. Differences start at day 0.

6.4 Oil, Gold and Equities

The commodity dimension separates the two shock types by the direction of crude. Over the (0, +5) window, Brent and WTI recorded cumulative abnormal returns

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of -14.32% and -13.14% after the tariff announcement, compared with $+27.63\%$ and $+34.55\%$ around the Hormuz onset. All four estimates are significant at the 1% level, and European equities fell in both episodes (Table 6.2). These patterns are consistent with weaker expected demand after the tariffs and concerns about current and future oil supply around Hormuz. The two benchmarks agree in sign and broad magnitude, so the result does not depend on the choice of crude reference. Nor does it depend on the currency of denomination. Re-expressed in euros, Brent falls 15.99% on the tariff shock and rises 29.21% on Hormuz (Table 6.2), so the oil moves are genuine commodity repricings, not the dollar moving in the denominator (Section 6.5 returns to this). The oil response is also the channel behind the one clean cross-sectional sort in Table 6.2, the underperformance of the oil exporters when the tariff shock depresses crude, while under Hormuz the two oil baskets fall together and the spread stays insignificant.

Gold tells a subtler story than the “universal haven” a first intuition would suggest. Rather than rising around every shock, gold in dollars tracks the inverse of the dollar across the two treatment shocks. It gains when the tariff shock weakens the dollar ($+6.07\%$ at ten days, and $+8.85\%$ in the ten days from the tariff pause) and falls when the Hormuz shock strengthens it (-8.74% at ten days, deepening to -22.30% at twenty, and -5.97% and -19.09% in euros). Re-expressing gold in euros shows how much the tariff-window gain depends on the currency in which it is measured. Over the ten days from Liberation Day gold gains $+0.21\%$ in euros, insignificant, against $+6.07\%$ in dollars. By construction the two figures differ by the euro’s own abnormal return against the dollar over the same days, $+5.86\%$, significant at the 1% level. Gold therefore rose against the dollar by about as much as the euro did and essentially not against the euro. This does not show that the dollar’s fall produced the gain. The euro was the second-strongest of the 32 currencies over these ten days while the broad dollar index fell 3.00% , so measured against a broad set of currencies about half of gold’s dollar gain remains. The gain from the tariff pause depends less on the currency, $+5.42\%$ in euros against $+8.85\%$ in dollars. What the euro series shows is that gold’s tariff-window gain is a gain against the dollar, shared with the currencies that absorbed the demand for safety (Section 6.1), and not a rise of gold against currencies in general. The external control behaves differently. Around the Ukraine invasion gold rises with the dollar and does so in euros as well ($+6.71\%$ at ten days, significant at the 1% level), a gain that holds in both currencies. Gold is thus not an unconditional substitute for the dollar but a hedge against it. For the two treatment shocks it moves opposite the dollar, reinforcing that the dollar’s own

conditional response, and not a generic flight to safety, is the organising variable of this chapter. Only for the external shock do both havens bid at once. One caveat tempers the magnitudes. Gold roughly tripled over the sample and its price peaked in late January 2026, days before the Hormuz estimation window closes, so the constant-mean baseline is unusually sensitive and the long-horizon gold CARs (e.g. Hormuz at twenty days) overstate the move. The sign is the robust feature, not the level.

Equities corroborate the risk-off reading without the dollar's conditionality. The Euro Stoxx 50 falls around every shock over the five-day window, by 14.58% on Liberation Day, 8.11% around Hormuz and 5.96% after the Ukraine invasion, all significant at the 5% level or better, and by an insignificant 3.05% around the twelve-day war, so European risk assets sell off whether the shock is a trade war, an oil-supply scare, or an external invasion. The contrast with the dollar is the point. European equities are an unconditional casualty of geopolitical and policy shocks, whereas the dollar's sign flips with the type of shock. The safe-haven question is therefore specific to the dollar, not a generic property of risk assets.

6.5 Robustness

Beyond the literature-standard basket panel of Table 6.2 and the interaction regression of Section 6.6, four further checks support the readings above. The first check uses longer windows. Because many things move daily prices at once, the ten- and twenty-day cumulations are not treated as independent identification of the event effect. Read as durability checks, they confirm that both treatment shocks' headline signs extend rather than reverse (-3.00% and -3.98% for the tariff shock, $+2.10\%$ and $+3.25\%$ for Hormuz), and the fifty-day horizon (Table 6.1) then separates a level re-rating of the dollar (tariffs, deepening to -6.34%) from a transitory risk premium (Hormuz, largely reverting, to -0.13% on the fifty-day estimation baseline and to $+1.07\%$ when read on the same $[-140, -21]$ baseline as the shorter horizons, with about two thirds of the twenty-day peak given back). The +50-day figures (e.g. Figure 6.4) plot the dollar, crude and the two oil-exposure baskets in separate panels, the three currency panels on one scale and crude on its own, and mark neighbouring sub-events, making both the persistence and the ceasefire reversal visible. The second check compares Brent with WTI. The crude responses share their sign and approximate size in every event, with WTI overshooting Brent materially only in the Hormuz spike (Table 6.2), so conclusions are invariant to the benchmark. The

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third check changes the denomination. Re-expressing the USD-priced commodities in euros with equation (5.1) moves every five-day cell (Table 6.2, bottom panel) by the euro's own abnormal return against the dollar, between 0.8 and 2.0 percentage points. Against crude responses of 11 to 28% this leaves sign and size essentially unchanged, so the crude moves are commodity repricings that no choice of currency undoes. For gold the shift is of the same order as the five-day response itself, which is why Section 6.4 reads gold in both currencies and rests its reading on the ten-day horizon. The fourth check, the market model of equation (5.5) with the equal-weighted dollar basket as market return, reported for the currency baskets in the Appendix, leaves the safe-haven ordering under both treatment shocks unchanged. It also sharpens the reading. Once the common dollar movement is removed, the safe basket beats the risky basket on Liberation Day (+1.31% against -0.33% over five days) and under Hormuz (+0.67% against -0.73%) alike, so the risk-off sorting among currencies is unconditional and what flips with the type of shock is whether the dollar takes part in it.

A final check concerns inference rather than the point estimates. The standard errors reported in this chapter follow the constant-mean convention of [MacKinlay \(1997\)](#). The residual variance is estimated on the pre-event window and the sampling error of the estimated normal return $\hat{\mu}$ of equation (5.2) is left out, so

$$\text{Var}(\text{CAR}) = L\hat{\sigma}^2.$$

Two departures from that convention both widen the interval. Carrying the estimation error of $\hat{\mu}$ through gives

$$\text{Var}(\text{CAR}) = L\hat{\sigma}^2(1 + L/T),$$

with $T = 120$ estimation days, an inflation of the standard error of 0.83% at $L = 2$, 8.40% at $L = 21$ and 19.37% at $L = 51$. And the event windows are visibly more volatile than the estimation windows that benchmark them. For the broad dollar the event-window standard deviation is 1.99 times its estimation-window value around Ukraine, 1.72 around Liberation Day and 1.65 around Hormuz, and trimming the two largest days of each window still leaves a ratio of 1.31 to 1.50, so this is sustained volatility rather than the announcement jump alone. A variance estimated before the event therefore understates the variance during it. Table 6.4 recomputes the headline contrast, Liberation Day – Hormuz at twenty days, under each departure and under both. The point estimate never moves and the contrast rejects at the 5% level under

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every treatment. What the corrections cost is the 1% starring, not the finding. Since the same widening applies to every cell, the stars printed at the ten- and twenty-day horizons are read as an upper bound on the strength of the evidence rather than as its measure, which is why inference rests on the short windows (Section 5.2).

Table 6.4: Sensitivity of the headline cross-event contrast to the variance estimate

Variance used in the standard error	ΔCAR (pp)	s.e. (pp)	t	Rejects at
As reported: estimation-window $\hat{\sigma}$, $L\hat{\sigma}^2$	−7.24	1.76	−4.11	1%
+ estimation error in $\hat{\mu}$: $L\hat{\sigma}^2(1 + L/T)$	−7.24	1.91	−3.79	1%
Event-window $\hat{\sigma}$ instead	−7.24	2.99	−2.42	5%
Both corrections	−7.24	3.24	−2.23	5%

Notes. Liberation Day – Hormuz crisis, broad nominal U.S. dollar index, $\text{CAR}(0, +20)$, constant-mean model, the headline cell of Table 6.3. $L = 21$, $T = 120$. The first row is the specification reported throughout, following MacKinlay (1997). The second adds the sampling error of the estimated normal return. The third replaces the pre-event residual standard deviation with the one measured inside the event window. The point estimate is identical in every row. Only the standard error moves.

Two limitations bound the long-horizon results. The ± 50 -day windows of the two 2025 events overlap, since the twelve-day war begins 52 trading days after Liberation Day and its own window reaches back into Liberation Day’s tail, so their long-horizon CARs are not fully independent. The persistence figures are therefore read event by event, while the formal cross-event difference tests (Table 6.3) use horizons from one to twenty days, at which the windows of the compared pairs do not overlap. A related caveat applies to the twelve-day war itself. Its estimation window contains the Liberation Day crash, which distorts the benchmark in both moments. The 41 trading days of the Liberation Day event window drag the estimated mean down (from +0.014 to −0.027 percent per day), mechanically raising the war’s measured abnormal returns. On a benchmark estimated without those days its twenty-day dollar CAR is −0.50 rather than +0.35 percent. Those days also inflate the estimated standard deviation, biasing the tests toward insignificance. The war’s muted response is therefore read as descriptive context rather than as a positive finding, with sign as well as significance uncertain. For the same reason the war is not paired with Liberation Day in Table 6.3. Its benchmark is estimated on days that include the whole Liberation Day event window, so Liberation Day’s abnormal returns would enter the difference twice, once directly and once through the war’s benchmark, and the printed gap would be overstated. And two further Gulf events are not analysed. The estimation window of the May 2026 U.S. strikes runs from November 2025 to late April 2026 and therefore covers the Hormuz crisis itself, so their abnormal returns

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would be measured against a benchmark that the episode under study has already inflated. The memorandum of understanding of 17 June 2026 falls too close to the sample end of 30 June 2026 to allow a full event window. The Hormuz window also contains one policy event before day 0. The Supreme Court’s ruling of 20 February 2026 that the International Emergency Economic Powers Act does not authorise the President to impose tariffs (Section 4.1) falls on day -6 , after the estimation window closes on 30 January 2026 and before the first headline window opens on day -1 , so it enters neither the benchmark nor any tabulated CAR. The dollar’s abnormal returns sum to -0.25% over the six trading days from the ruling to day -1 ($t = -0.45$), against $+1.39\%$ over the two-day announcement window.

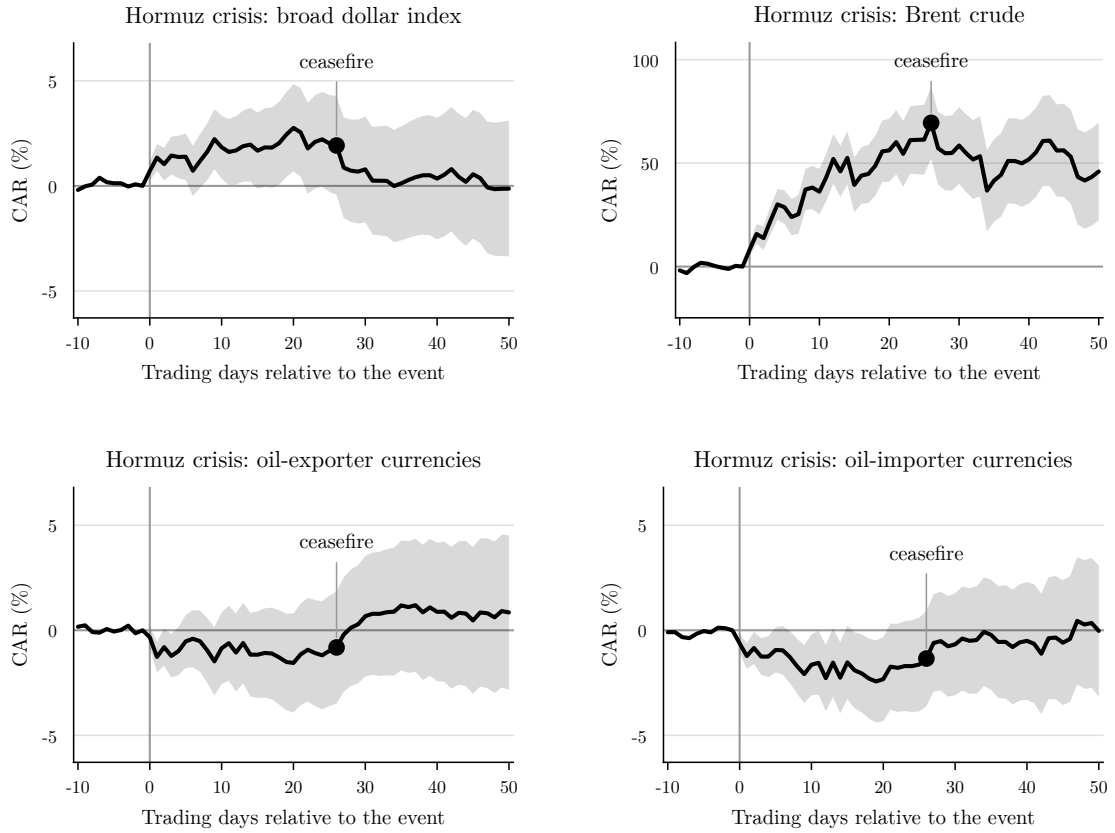


Figure 6.4: Hormuz crisis, -10 to $+50$ trading days, estimation window $[-170, -51]$ so that it does not overlap the wider event window. Construction as in Figure 6.1. The 7 April two-week ceasefire is marked.

6.6 Regression Results

The event study above shows that the dollar moved in opposite directions across the two shock types. This section asks a narrower question. Did the dollar's safe-haven property, appreciating when global risk rises, hold or break in each episode? The dummy-interaction regression of equation (5.6) addresses it with ordinary least squares and Newey–West standard errors, estimated in four versions that differ only in the measure of risk.

Table 6.5: Did the dollar's safe-haven reflex survive each shock? (OLS, HAC standard errors)

	(A) VIX	(B) VIX + GPR	(C) VXY	(D) GPR + TPU
Δ VIX (normal)	+0.069*** (+7.01)	+0.070*** (+7.06)	—	—
× Tariff	−0.071*** (−2.58)	−0.067*** (−2.70)	—	—
× Hormuz	+0.057 (+1.31)	+0.062 (+1.30)	—	—
× Ukraine	+0.165*** (+4.84)	+0.168*** (+3.97)	—	—
Δ GPR (normal)	—	+0.013** (+2.04)	—	+0.011* (+1.73)
× Tariff	—	+0.154* (+1.82)	—	+0.180* (+1.81)
× Hormuz	—	+0.002 (+0.04)	—	−0.011 (−0.25)
× Ukraine	—	0.000 (0.00)	—	−0.025 (−0.46)
Δ VXY (normal)	—	—	+0.107*** (+6.93)	—
× Tariff	—	—	−0.127*** (−3.20)	—
× Hormuz	—	—	+0.161*** (+3.89)	—
× Ukraine	—	—	+0.117*** (+4.29)	—
Δ TPU (normal)	—	—	—	−0.003 (−0.47)
× Tariff	—	—	—	−0.022 (−1.10)
× Hormuz	—	—	—	+0.051 (+0.54)
× Ukraine	—	—	—	+0.032 (+0.11)
Window dummies	Yes	Yes	Yes	Yes
N	1954	1954	1954	1954
R^2	0.057	0.061	0.137	0.009

Notes. OLS of the daily broad-dollar return (percent) on daily risk shocks, their interactions with crisis-window dummies, and the window (level) dummies. Newey–West (HAC, 5 lags) standard errors are used, with t -statistics in parentheses. Each risk shock is standardized to unit variance, so coefficients read as the percent dollar move on a one-standard-deviation risk-off day. Crisis windows run from the event day to +20 trading days. VIX: CBOE (FRED VIXCLS); GPR: [Caldara and Iacoviello \(2022\)](#) daily index; VXY: J.P. Morgan VXY Global FX implied volatility (LSEG .JPMVXYGL, mid); TPU: [Caldara et al. \(2020\)](#) daily index, current vintage (downloaded 23 July 2026). A positive slope is safe-haven behaviour (risk up $\rightarrow$ dollar up). A negative interaction weakens it. */**/***: 10/5/1%.

The three specifications using market-priced risk measures, A to C in Table 6.5, reach the same verdict, while the news-based specification D does not, as set out at the end of this section. On normal days the dollar is a safe haven. In the baseline

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VIX specification a one-standard-deviation risk-off day lifts the broad dollar by 0.069% ($t = 7.01$). During the tariff window this reflex disappears, the interaction is equal and opposite (-0.071 , $t = -2.58$) and the within-window sensitivity is essentially zero. Within this window the dollar showed no haven response, and this is the one episode in which the United States was the source of the shock. During the Hormuz window the interaction is positive but insignificant ($+0.057$, $t = 1.31$), leaving the normal relationship intact, and during the Ukraine window the reflex strengthens ($+0.165$, $t = 4.84$). Controlling for geopolitical news changes none of this. In the joint specification the VIX terms are essentially unchanged, the GPR index enters with a small positive normal-day slope, and its interactions are at most marginally significant. The currency-market measure sharpens the result. Implied dollar exchange-rate volatility, the VXY index, explains daily dollar returns more than twice as well as the VIX ($R^2 = 0.137$ against 0.057), and each of its window interactions carries the sign the safe-haven reading predicts and is significant on the table's standard errors, positive on normal days ($+0.107$, $t = 6.93$), erased during the tariff window (interaction -0.127 , $t = -3.20$, so the within-window slope is again near zero), and reinforced during the Hormuz (interaction $+0.161$, $t = 3.89$) and Ukraine ($+0.117$, $t = 4.29$) windows. The one measure priced in the currency market itself thus registers both the breakdown and the survival of the reflex within a single specification. The news-count pair, by contrast, has little daily explanatory power ($R^2 = 0.009$), with only the GPR interaction in the tariff window marginally positive ($+0.180$, $t = 1.81$), an honest negative for daily news-based measures. The estimated window dummies, not tabulated, echo the event study, with the dollar's average drift negative in the tariff window and positive in the Hormuz window in all four specifications, and positive in the Ukraine window in three of the four.

The design limits the weight this evidence can carry. Each interaction is identified from the twenty-one trading days of its window, so its standard error depends on how it is computed, while the coefficient itself does not. For the tariff interaction of the baseline specification the Newey–West t -statistic of -2.58 becomes -1.41 with the heteroskedasticity-robust standard errors of [White \(1980\)](#), which make no correction for serial correlation, and -1.06 with the leverage-adjusted form of those standard errors (HC3) proposed by [MacKinnon and White \(1985\)](#), and the VXY specification's -3.20 becomes -2.02 and -1.68 . The point estimate also moves with single days. Removing 3 April 2025 from the tariff window shrinks the baseline interaction to -0.044 and removing 4 April enlarges it to -0.126 , although the within-window slope stays between -0.06 and $+0.03$ in every such sample, well below the normal-day

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slope of +0.069. The stars in Table 6.5 are therefore not the evidence. The evidence is the pattern of the point estimates, a risk sensitivity near zero inside the tariff window and clearly positive inside the two military windows, and that pattern is the event study's.

The regression thus supports the event study's conclusion. In the three specifications using market-priced risk measures the point estimates show the dollar's safe-haven function breaking down in the U.S. trade-policy shock and remaining intact or strengthened in the military and oil-supply shocks. The news-based specification, which explains little of the daily variation, does not reproduce the tariff-window breakdown. Its GPR interaction is positive and significant only at the 10% level and its TPU interaction is insignificant, so the verdict rests on the specifications with market-priced measures.

7 Discussion

This chapter draws the event-study and regression results together, relates them to the literature, and sets out their implications and limits.

7.1 Findings in Light of the Literature

The results speak directly to the six literatures set out in Chapter 2. The sharpest verdict is for the dollar-erosion strand. [Kamin \(2025\)](#) and [Jiang et al. \(2026\)](#) read the 2025 tariff episode as evidence that the dollar has stopped behaving as a safe haven, and [Hassan et al. \(2025\)](#) supply the theory. A trade war that isolates U.S. goods markets erodes the currency's safety premium. The findings here both confirm and bound that claim. They confirm it for the tariff shock, when the dollar depreciates and its risk sensitivity collapses to zero, but they bound it by showing that the erosion is not a general weakening, since the very same dollar appreciates around the contemporaneous Hormuz shock. What the dollar-erosion papers identify as a regime change is, on this evidence, specific to U.S.-originated economic-policy shocks rather than a property of the post-2025 dollar as such. Imposing the tariffs took the broad dollar index down 1.44% over the two-day announcement window, while the Supreme Court ruling of 20 February 2026 that removed their legal basis moved it by only -0.25% over six trading days (Section 6.5). This asymmetry fits the lasting re-rating of Section 6.1 better than a premium that returns once the policy is withdrawn, although the ruling followed earlier court decisions against the tariffs and left other tariffs in place (Section 4.1), so markets may have priced much of it in advance.

For the safe-haven tradition ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#); [De Bock and de Carvalho Filho 2015](#)) the evidence is corroborative. The dollar performs as that literature predicts around both military/oil-supply shocks, appreciating as risk rises, and the formal difference test separates the U.S.-involved Hormuz response from the external Ukraine benchmark only on impact and marginally at twenty days, so the haven bid survives even U.S. involvement when the shock is a security one. The conditional, crisis-contingent definition of [Baur and Lucey \(2010\)](#) is thus vindicated for the dollar, whose haven status is real but state-dependent. That same framing organises the gold result. Gold hedges against the dollar rather than serving as a universal safe haven. It bids when the tariff shock weakens the dollar and falls when the Hormuz shock strengthens it, rising alongside the dollar only for the external Ukraine shock. This is exactly the state-dependence Baur and

Lucey’s distinction implies.

The commodity-currency literature ([Y.-C. Chen and Rogoff 2003](#); [Kilian 2009](#)) is borne out only in part. At the five- to twenty-day horizons the exporter–importer spread carries the sign of the crude move in the three main episodes, but the clearest separation occurs after the tariff shock, the Hormuz spread remains insignificant, and Ukraine provides weaker evidence only at twenty days (Section 7.3). Finally, the geopolitical-risk literature ([Caldara and Iacoviello 2022](#)) supplies both the measurement and a mechanism. The within-crisis ceasefire reversal, which unwinds the dollar premium as the threat recedes, is the threat-versus-realisation distinction Caldara and Iacoviello emphasise, mapped onto the exchange rate.

7.2 The Dollar’s Safe Haven Is Conditional on Shock Type

The central finding is that the dollar’s safe-haven response is conditioned by the type of shock, not merely by who causes it. The dollar appreciated around both military/oil-supply shocks, the external Ukraine invasion and the U.S.-involved Hormuz crisis, and the formal difference test cannot distinguish the two responses at five and ten days, even though the United States was a belligerent in one and merely witnessed the other. Yet it depreciated around the one purely economic-policy shock it created, the Liberation Day tariffs. Origin alone does not explain the pattern. The nature of the shock does. This refines the “dollar-erosion” reading ([Kamin 2025](#); [Jiang et al. 2026](#)). The safe-haven function is not weakening in general but is specifically suspended when the United States is itself the source of an economic-policy shock, while it survives intact, despite the subsequent reversion, for the security shocks that the safe-haven tradition has always emphasised ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#)).

The regression supports this reading. Estimated over the full sample, the dollar’s sensitivity to risk is significantly positive on normal days, the textbook safe-haven reflex, and the point estimates show it falling to about zero inside the tariff window while remaining positive inside the Hormuz and Ukraine windows. The pattern is thus not only a difference in the level of the dollar across episodes but also a difference in its relationship with risk, although the window estimates rest on few observations and their significance depends on the standard error used (Section 6.6).

Read against the hypotheses of Section 3.4, the evidence adjudicates all three. The first hypothesis is supported. The dollar appreciated around both military/oil-supply

shocks and depreciated around the trade-policy shock, and the difference-in-CARs test separates the tariff episode from both others while leaving Hormuz and Ukraine indistinguishable at the five- and ten-day horizons, with only a difference of intensity on impact and a marginal gap at twenty days. The second hypothesis receives mixed evidence rather than robust support. In the U.S.-originated tariff episode gold's ten-day cumulative abnormal return is positive in dollars (+6.07%, significant at the 5% level) but small and insignificant in euros (+0.21%), and its five-day dollar-denominated estimate is negative and insignificant (−1.54%), so the prediction is met at the ten-day dollar-denominated horizon and the result is sensitive to horizon and denomination. Gold's negative cumulative abnormal returns in the Hormuz crisis (−22.30% at twenty days, significant at the 1% level) further weaken its interpretation as an origin-blind safe-haven benchmark (Section 7.3). The third hypothesis is partly supported. At the five- to twenty-day horizons the exporter–importer spread carries the sign of the crude response in the three main episodes, negative under the tariff shock and positive under Hormuz and Ukraine, but the separation is significant at the 1% level only for the tariff shock and, more weakly, at twenty days for Ukraine (Section 7.3).

7.3 Gold and the Oil Channel

Gold does not behave as a universal safe haven, an asset that is uncorrelated or negatively correlated with risky assets in times of market stress in the sense of [Baur and Lucey \(2010\)](#), whatever the source of that stress. Instead it tracks the inverse of the dollar across the two U.S.-linked shocks, rising when the tariff shock weakened the dollar, falling when the Hormuz shock strengthened it, and bids alongside the dollar only for the external Ukraine shock. Gold is thus better described as a hedge against the dollar than as a substitute for it, which reinforces, rather than competes with, the chapter's main message, that the dollar's own conditional move is the organising variable and gold prices it from the other side.

The oil channel sorts the currencies as the commodity-currency literature predicts only in part. The clearest exporter–importer separation came after the tariff announcement, when crude fell, with an exporter-minus-importer spread of −2.80 percentage points over days 0 to 10, significant at the 1% level. Under Hormuz, when crude surged, the spread was positive but insignificant at five, ten and twenty days, and around the Ukraine invasion it reached +4.38 percentage points at twenty days, significant only at the 10% level. The pattern is consistent with the finding of

H. Chen et al. (2016) that dollar exchange rates respond to demand-driven rather than supply-driven oil moves, although their monthly structural model and this daily event study identify different objects, and the overlap between the oil-importer and safe baskets (Section 6.1) prevents the spread from isolating oil exposure from safe-haven characteristics. For currencies such as the yen, safe-haven demand and higher oil-import costs may exert opposing pressures during an oil-supply shock.

7.4 Policy Implications

Three implications follow. For reserve managers and central banks, the dollar's safe-haven status cannot be treated as unconditional, because hedging programmes calibrated on the assumption that the dollar always rallies in risk-off episodes will misfire precisely when the shock originates in U.S. economic policy. For portfolio managers, diversification should be thought of across shock types, not only across assets. In the tariff episode the flight to quality went to competing-haven currencies rather than to the dollar, and gold gained against the dollar by about as much as those currencies did but not against them. For the broader debate on the dollar's reserve role (Obstfeld and Zhou 2023; Jiang et al. 2026), the evidence is a caution against over-reading any single episode. The 2025 tariff shock did dent the dollar, but the 2026 security shock showed the haven bid still firmly in place.

7.5 Limitations

Several caveats bound the results. Two Gulf events, the U.S. strikes of 25 May 2026 and the memorandum of understanding of 17 June 2026, are not analysed. The estimation window of the first contains 40 of its 120 trading days on or after the 2 March onset of the Hormuz crisis, so its benchmark would be contaminated, and the second lies too close to the sample end of 30 June 2026 for the longer windows (Section 6.5). The classification of the Hormuz crisis as a U.S.-involved shock is a snapshot taken at its onset, when the United States and Israel struck jointly. Israel struck alongside the United States until the ceasefire of 8 April 2026, exchanged fire with Iran once more on 8 June and sat out the fighting between the United States and Iran in July (Section 4.2). If the war continues as an American one, the episode drifts over a long enough horizon toward a U.S.-originated shock and blurs the distinction between the origin and the nature of a shock on which the design rests, although nothing after the sample end of 30 June 2026 enters the estimates. The

regression’s crisis windows are defined around the events rather than by a structural break test, and the VIX–dollar link it measures is a reduced-form association, not a causal channel. Its interaction terms rest on twenty-one observations each, and the dollar’s sensitivity to risk is not constant over the rest of the sample either. When the tariff window is moved to each of the 1,811 other twenty-one-day stretches of the sample that do not overlap the modelled windows, the interaction is significant at the 5% level on the reported standard errors in 28% of them for the VIX and 44% for the VXY, and in 12% and 24% on heteroskedasticity-robust standard errors. A significant interaction in a single window is therefore weak evidence on its own, which is why the regression is read as supporting evidence for the event study rather than as a separate test (Section 6.6). Gold’s exceptional 2025–26 trend makes its constant-mean CARs sensitive in level, so they are read for sign rather than magnitude. As a single-country event study around a small number of recent, partly overlapping episodes, the design trades breadth for the clean U.S.-versus-external contrast, and without forward/basis data beyond the one-month tenor, covered-interest-parity deviations are left for future work.

8 Conclusion

This thesis asked whether the U.S. dollar’s safe-haven response depends on the type of shock that triggers a crisis, even when the United States is implicated in the shock itself. Two recent episodes allowed a direct comparison. The 2025 Liberation Day tariffs were a purely U.S. economic-policy shock and the 2026 Iran/Hormuz crisis a geopolitical oil-supply shock, with the 2022 Russian invasion of Ukraine as an external control.

The evidence gives a clear answer. The two shock types produced a mirror image. The dollar depreciated around the tariff shock by 3.98 percent on the broad index within twenty trading days, while it appreciated around the Hormuz crisis and around Ukraine, and a formal difference test confirms the tariff response is significantly distinct from both security episodes, which are themselves statistically indistinguishable from one another at the five- and ten-day horizons. The dollar’s haven status is therefore conditional on the nature of the shock rather than on its origin in the three episodes studied. It survives a U.S.-involved security shock but inverts for a U.S.-originated economic-policy shock. A complementary regression supports the same reading in its point estimates. The dollar’s sensitivity to risk, significantly positive on normal days, falls to about zero in the tariff window and holds through the Hormuz and Ukraine episodes. Two further results round out the picture. Gold acts as a hedge against the dollar rather than as a universal safe haven, and the clearest exporter–importer separation occurs after the tariff shock, when crude fell, while the Hormuz spread remains insignificant and Ukraine provides weaker evidence at twenty days.

The thesis contributes a comparison of the Liberation Day and Iran/Hormuz episodes within one consistent design. It reads the dollar’s response alongside the currencies most exposed to risk appetite and crude prices and alongside gold, in dollars and in euros, and so connects the safe-haven and commodity-currency literatures without treating the dollar’s movement in isolation.

Several extensions suggest themselves. As the geopolitical situation continues to evolve, the sample can be extended so that the most recent episodes acquire full post-event windows. Richer forward and basis data would allow covered-interest-parity deviations to be brought into the analysis, and intraday data would tighten the identification around the announcement windows. A dynamic conditional-correlation (DCC-GARCH) model of the dollar–VIX relationship would test the safe-haven-breakdown hypothesis directly and is the natural next layer. Finally, a formal model

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of shock-dependent safe-haven demand would give the empirical pattern documented here a theoretical foundation. In such a model, the currency of the shock's originator would lose its haven premium for policy shocks but not for security shocks.

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Appendix

1 Currency Universe and Classification

Table 1 lists the 32 currencies in the panel with their carry-sorted bucket (low/middle/high forward-discount tercile, pegged and managed currencies excluded from the sort) and their oil-trade role. The literature-standard robustness baskets used in Chapter 6 are the yen, Swiss franc and euro (safe) against the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah (risky).

Table 1: Currency classification

Code	Currency	Carry bucket	Oil role
AUD	Australian dollar	middle	—
BRL	Brazilian real	no 1M forward	exporter
CAD	Canadian dollar	middle	exporter
CHF	Swiss franc	low (safe)	—
CLP	Chilean peso	no 1M forward	—
CNY	Chinese yuan	pegged/managed	—
COP	Colombian peso	no 1M forward	exporter
CZK	Czech koruna	middle	—
DKK	Danish krone	pegged (EUR)	—
EUR	Euro	low (safe)	—
GBP	Pound sterling	middle	—
HKD	Hong Kong dollar	pegged (USD)	—
HUF	Hungarian forint	high (risky)	—
IDR	Indonesian rupiah	high (risky)	—
ILS	Israeli shekel	low (safe)	—
INR	Indian rupee	high (risky)	importer
JPY	Japanese yen	low (safe)	importer
KRW	South Korean won	middle	importer
KWD	Kuwaiti dinar	pegged	exporter
MXN	Mexican peso	high (risky)	exporter
MYR	Malaysian ringgit	middle	—
NOK	Norwegian krone	middle	exporter
NZD	New Zealand dollar	middle	—
PEN	Peruvian sol	no 1M forward	—
PLN	Polish zloty	high (risky)	—
SAR	Saudi riyal	pegged	exporter
SEK	Swedish krona	low (safe)	—
SGD	Singapore dollar	middle	—
THB	Thai baht	low (safe)	importer
TRY	Turkish lira	high (risky)	importer
TWD	Taiwan dollar	low (safe)	importer
ZAR	South African rand	high (risky)	—

Notes. Carry buckets are terciles of the average one-month forward discount (Section 5.1.3). Pegged/managed currencies and the four currencies without a clean one-month forward (BRL, CLP, COP, PEN) are excluded from the carry sort. Oil roles use observed crude oil trade over 2019–2024 (Section 5.1.3). Partner reports retrieved from the UN Comtrade API on 15 September 2026 supplement 31 of 33 missing country-flow-year observations across India, Japan, Kuwait, Mexico, Taiwan and Saudi Arabia. Kuwait’s imports in 2019 and 2023 remain unobserved, not zero. For Türkiye, partner reports imply imports of 28.58 billion dollars and exports of 9.56 billion dollars, an observed import surplus of 19.02 billion dollars. Its import mirror contains no Iraqi reports for 2019–2024 or Russian reports for 2022–2024. Complete partner coverage cannot be established, and mirror values may differ from national totals. All observed balances support the stated roles, conditional on the remaining coverage gaps.

2 Robustness

Market model. Alongside the constant-mean specification, the event study computes market-model abnormal returns, using the equal-weighted dollar basket as the market return (`Code/event_study/03_event_study.py`, method `market_model`). Table 2 reports the basket-level results. The safe-haven ordering under both treatment shocks is unchanged relative to the constant-mean results in Chapter 6.

Table 2: Market-model CARs of the currency baskets

	Ukraine		Liberation Day		12-day war		Hormuz	
	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)
Safe currencies	-0.31	-1.73**	+1.31***	+1.17*	-0.13	+0.21	+0.67**	+0.67
Risky currencies	-0.89	-0.21	-0.33	-0.48	-0.26	+0.31	-0.73***	-1.20***
Oil exporters	+1.26***	+2.29***	-0.62	-0.77	+0.34	+0.10	+0.61	+1.23*
Oil importers	+0.51	-1.97	+1.00***	+0.95	-0.49	-0.82	-0.005	-0.02
Carry (risky-safe)	-0.57	+1.52	-1.63***	-1.65	-0.13	+0.10	-1.40***	-1.87**

Notes. Cumulative abnormal returns of the currency baskets in percent. Abnormal returns are residuals from a regression of each basket on the equal-weighted dollar basket estimated over the same $[-140, -21]$ window (Section 5.2). The safe-haven ordering under both treatment shocks is unchanged. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Further event figures. Figure 1 shows the full portfolio response around the external Ukraine benchmark at the identification window. In the main text the Ukraine dollar path enters only through the cross-event differences of Figure 6.3. Figure 2 shows the same panels for the twelve-day war. Its estimation window contains the Liberation Day event window, so its paths and bands are read with the caveat of Section 6.5. Figure 3 extends the Liberation Day window to +50 days, plotted from -10 to +50. The tariff drift deepens rather than reverts, the mirror image of the transitory Hormuz premium in Figure 6.4.

Appendix

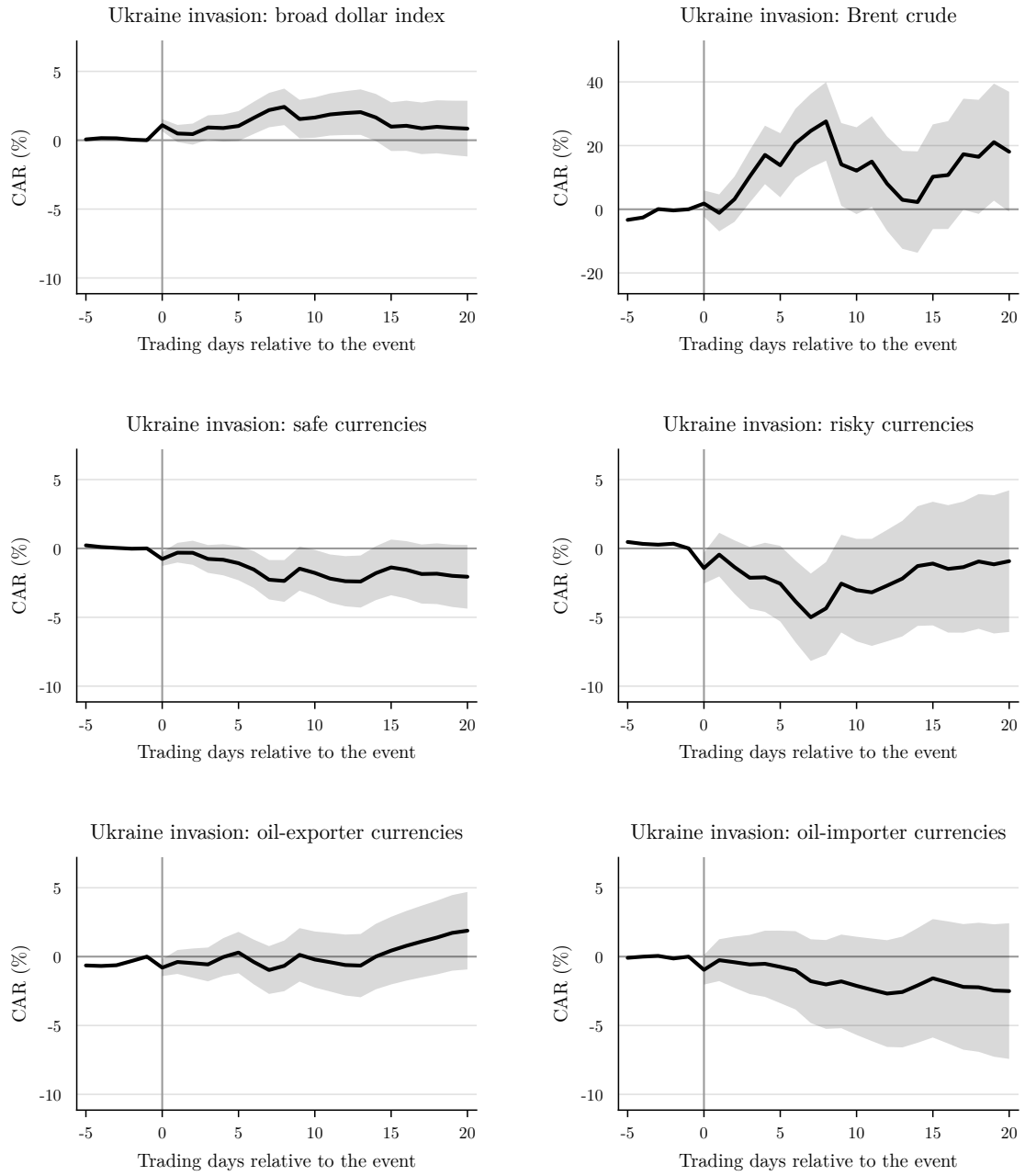


Figure 1: Ukraine invasion (24 February 2022): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1.

Appendix

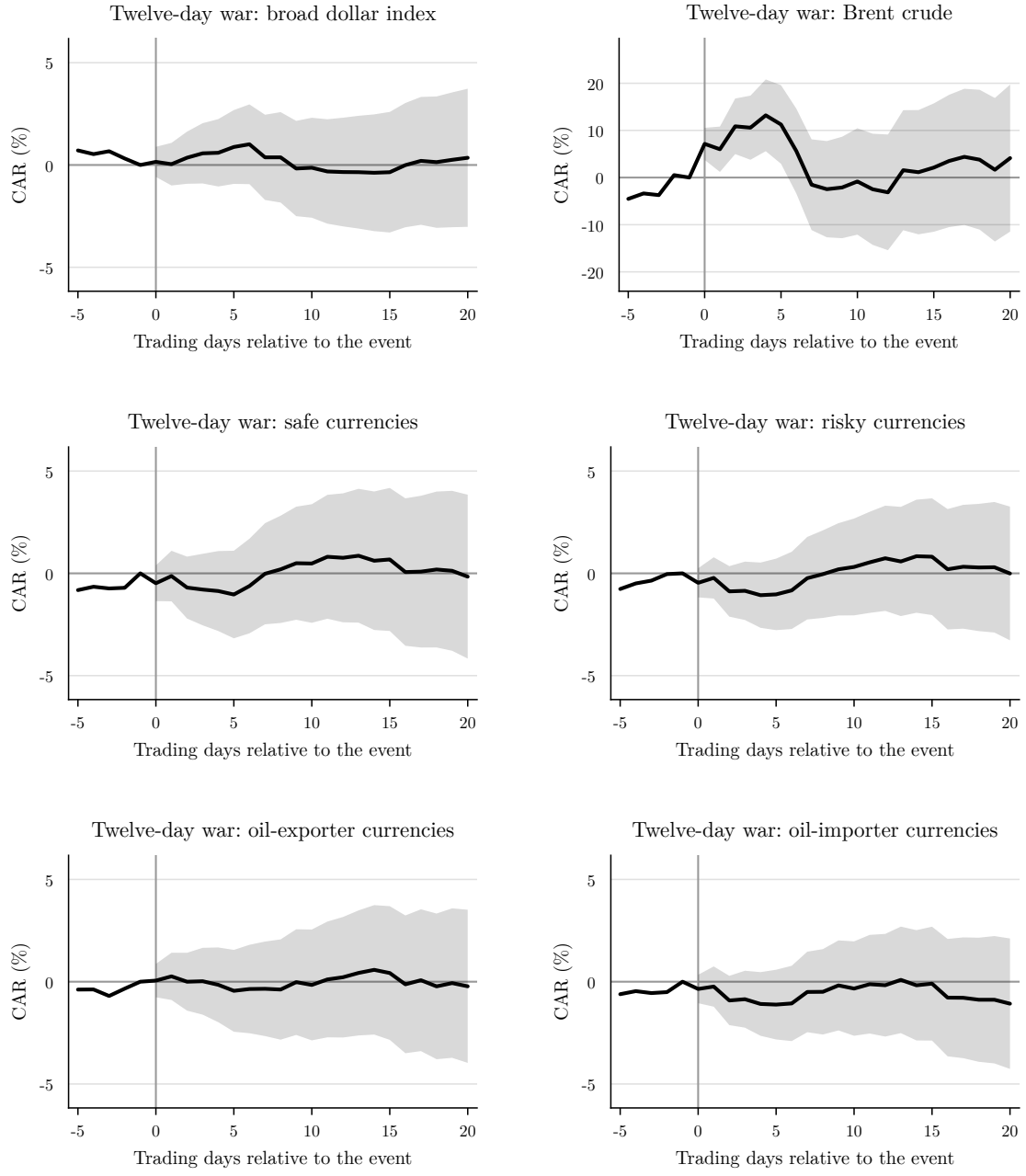


Figure 2: Twelve-day war (13 June 2025): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1. The estimation window contains the Liberation Day event window (Section 6.5).

Appendix

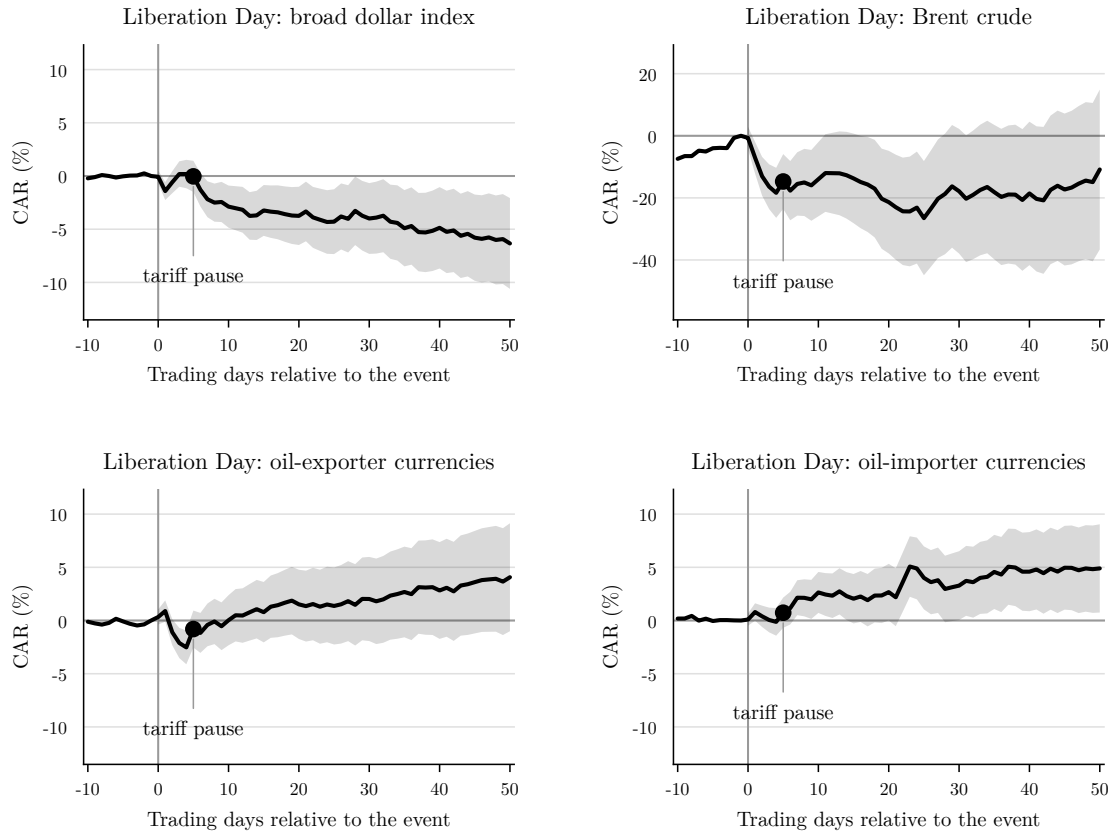


Figure 3: Liberation Day, -10 to $+50$ trading days, estimation window $[-170, -51]$. Construction as in Figure 6.1. The post-announcement dollar decline deepens over the horizon. The 9 April tariff pause (marked) interrupts but does not reverse it.

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